Fortuitous Q-cohomology and quantum lifting: master report

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0. Conventions

Spectral sequence

$$\mathbf{Q}=\sum_{n\geq 0}\hbar^n Q_n, \qquad \sum_{j=0}^{n} Q_j Q_{n-j}=0, \qquad E_1=H(Q_0)$$

$$\lambda=(\theta_1,\theta_2,\theta_3;n_+,n_-), \qquad \Theta=\theta_1+\theta_2+\theta_3, \qquad Y=\#\text{letters}, \qquad \mathsf{d}=(1,1,1;-1,-1)$$

$$e_I=\text{unit vector of }\theta_I, \qquad e_\pm=\text{unit vector of }n_\pm$$

$$Q_n:\ C^Y(\lambda)\to C^{Y+1-n}(\lambda-n\mathsf{d}), \qquad d_r:\ E_r^Y(\lambda)\to E_r^{Y+1-r}(\lambda-r\mathsf{d})$$

$$\text{outgoing } d_r:\ r\leq \min_I \theta_I, \qquad \text{incoming } d_r:\ r\leq \min(n_+,n_-)$$

$$d_2[\mathcal{O}_0]=[Q_2\mathcal{O}_0+Q_1\mathcal{O}_1], \qquad Q_0\mathcal{O}_1=-Q_1\mathcal{O}_0$$

$$\mathcal{O}(\hbar)=\sum_{r\geq 0}\hbar^r\mathcal{O}_r, \qquad Q_0\mathcal{O}_r=-R_r, \qquad R_r=\sum_{k=1}^{r}Q_k\mathcal{O}_{r-k}$$

$$\operatorname{rk} d_R=\left(\operatorname{rk} M_R-\operatorname{rk} M_{R-1}\right)-\left(\operatorname{rk} M'_R-\operatorname{rk} M'_{R-1}\right)$$

$$M_R=\Big[\ \underbrace{Q_k\mathcal{K}}_{\text{cocycles}}\ \Big|\ Q_{k-m}:C_m\to C'_k\ \Big]_{1\leq m\leq k\leq R}, \qquad M'_R=M_R\setminus\{Q_k\mathcal{K}\}$$

$$C_m=C^{Y-m}(\lambda-m\mathsf{d}), \qquad C'_k=C^{Y-k+1}(\lambda-k\mathsf{d})$$

Q_n on matrix letters

$$Q_nF=\mathrm{CENG}[n]\,\mu^{-n}\sum_{(i_1,\dots,i_{n+1})}\varepsilon\,A_nK_n\,F\big(\dots,\underbrace{[P,t_{R_1}]}_{i_1},\dots,\underbrace{[t^{R_1},t_{R_2}]}_{i_2},\dots,\underbrace{[t^{R_n},Q]}_{i_{n+1}},\dots\big)$$

$$\mu=-2, \qquad \mathrm{CENG}=(1,1,2,4,8), \qquad t^R=\kappa^{RS}t_S, \qquad \kappa_{RS}=\operatorname{tr}_F(t_Rt_S)$$

Charges

$$E=\tfrac12(\Theta+Y)+n_++n_-, \qquad R_I=\theta_I-\tfrac12(\Theta-Y), \qquad J_L=\tfrac12(\Theta-Y+n_++n_-), \qquad J_R=\tfrac12(n_+-n_-)$$

$$E=R_1+R_2+R_3+2J_L, \qquad 2(E+J_L)=2\Theta+3(n_++n_-)$$

$$(E,R_I,J_L,J_R)(\lambda+k\mathsf{d},\,Y+k)=(E,R_I,J_L,J_R)(\lambda,Y), \qquad (E;R_I;J_L,J_R)(\mathbf{Q})=(\tfrac12;\tfrac12,\tfrac12,\tfrac12;-\tfrac12,0)$$

twisted letter	physical letter	θ	jets	E	(R_1,R_2,R_3)	J_L	J_R
$(\gamma^I)_A$	$\bar\phi_I{}^A$	e_I	–	1	e_I	0	0
$(\beta_I)^A$	$-\frac{i}{\sqrt2}\psi_+^{IA}$	$(1,1,1)-e_I$	–	$\frac32$	$(\frac12,\frac12,\frac12)-e_I$	$\frac12$	0
$\delta^{AB}\mathsf{b}_B$	$\frac14 f_{++}^A$	$(1,1,1)$	–	2	$(0,0,0)$	1	0
$\partial_{+\dot a}\mathsf{c}^A$	$2\bar\lambda^A_{\dot a}$	0	$n_\pm=1$	$\frac32$	$(\frac12,\frac12,\frac12)$	0	$\pm\frac12$
$\partial_{+\dot a}$	$\nabla_{+\dot a}$	0	$n_\pm=1$	1	$(0,0,0)$	$\frac12$	$\pm\frac12$

Quantities

$$h=\dim H^Y(Q_0)\text{ on highest-weight cochains of }(\lambda,Y),\qquad g=\#\text{graviton classes}$$
$$\rho=\mathrm{rk}\big(\mathrm{res}_{\mathfrak h}:H^Y\to\text{Cartan restriction}\big)$$
$$0\to R_N\to C_\infty\xrightarrow{\pi_N}C_N\to0,\qquad f=h-\mathrm{rk}\,\pi_N^*$$
$$K(Y)=\dim\ker\big(H^Y(R_N)\to H^Y(C_\infty)\big)=f(Y-1)$$

Certification [$\mathbb{Q}$] exact over $\mathbb{Q}$; [p] ranks mod $p=2^{31}-1$; [p,p′] also $p'=2147483629$; [cond] $\mathrm{res}(R_r)=0$ at targets with $h=\rho$.
All orders $\mathbf{Q}\,\mathcal{O}(\hbar)=0$ (closed completion) $\wedge$ $\mathcal{O}_0\notin\mathrm{im}\,d_r\ \forall r$.

1. $\mathcal{N}=4$ Yang–Mills, $G=SU(N)$

1.1 Action and components

$$C^A(\theta)=ig\,\mathsf{c}^A+\theta_I\delta^{AB}(\gamma^I)_B-\tfrac12\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A+\frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B,\qquad Q_+^IC^A=\frac{\partial_LC^A}{\partial\theta_I}$$
$$A=1,\dots,N^2-1$$
$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\int d^3\theta\Big(\tfrac12\delta_{AB}C^A\bar\partial C^B+\tfrac{i}{3!}C_{ABC}C^AC^BC^C\Big)\Big]_{(0,2)},\qquad Q_0C^A=\tfrac{i}{2}C^A{}_{BC}C^BC^C$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)$$
$$+\sum_{I=1}^3\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+\tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$
$$(t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\qquad[t_B,t_C]=iC^A{}_{BC}t_A,\qquad\delta_{AB}=2\,\mathrm{tr}(t_At_B)$$
$$\mathsf{w}^A_{\dot a}\mapsto\tfrac12\partial_{+\dot a}\mathsf{c}^A,\qquad-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}=-\tfrac{1}{16}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^A)$$
$$\mathcal{I}=-\tfrac{g}{2}C^A{}_{BC}\mathsf{b}_A\mathsf{c}^B\mathsf{c}^C-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C(\beta_I)^B$$
$$+\tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}$$

components: $\underbrace{\mathsf{c}^A,\ \mathsf{b}_A}_{\text{vector}}$, $\underbrace{(\gamma^I)_A,\ (\beta_I)^A}_{I=1,2,3\text{ adjoint chirals}}$, C_{ABC} of $\mathfrak{su}(N)$; letters $X^{(r,s)}=\partial_{+\dot1}^r\partial_{+\dot2}^sX$, c only with $r+s\ge1$

1.2 Lie algebra realization

$SU(2)$	basis $(e,h,f)=(E_{12},\ E_{11}-E_{22},\ E_{21})$	$[e,h]=-2e,\ \ [e,f]=h,\ \ [h,f]=-2f$
$SU(3)$	$(e_1,e_2,e_3,h_1,h_2,f_1,f_2,f_3)$	$=(E_{12},E_{23},E_{13},E_{11}-E_{22},E_{22}-E_{33},E_{21},E_{32},E_{31})$
	letter $A=\sum_X A^Xt_X,\ \ [t_Y,t_Z]=C^X{}_{YZ}t_X$	$D_ZA^X=C^X{}_{ZY}A^Y,\ \ \kappa_{XY}=\mathrm{tr}(t_Xt_Y)$
	singlets	$\mathrm{wt}=0\ \cap\ \ker D_{f_1}\cap\ker D_{f_2}\quad(SU(2)\text{: }\mathrm{wt}=0\cap\ker D_f)$
	charge conjugation	$Ct_X=-t_X^{\mathsf T},\quad C=\pm\text{ split }\frac12(1\pm C)$
$SU(N)$	fundamental $N\times N$, tr $=0$	$\{E_{ij}\}_{i\ne j}\cup\{E_{ii}-E_{i+1,i+1}\},\ \ t^R=\kappa^{RS}t_S$
	Casimir tensor	$\sum_R(t_R)_{ab}(t^R)_{cd}=\delta_{ad}\delta_{bc}-\frac1N\delta_{ab}\delta_{cd},\quad C_F=\frac{N^2-1}{N}$
	probes	$x\in\mathfrak{sl}_N(\mathbb{F}_p)$: N^2-1 random entries, $x_{NN}=-\sum_{i<N}x_{ii}$; $\mathrm{Tr}\,\mathbf1=N$
	trace words ($N\ge4$)	$C_\infty=\mathbb{C}[\mathrm{tr}(w)]_{\lvert w\rvert\ge2},\quad R_N=\ker\big(\pi_N:C_\infty\to C_N\big)$ (trace identities)

1.3 Highest-weight algorithm

(a) $SU(2)$, $SU(3)$: bosonic highest weights

$$G_{\mathrm{HW}} = GL(3)_\theta \times SL(2)_{\dot a} \ \text{(letter-wise)}, \qquad [G_{\mathrm{HW}}, Q_n] = 0$$

$$\theta(\gamma^I) = e_I, \quad \theta(\beta_I) = (1,1,1) - e_I, \quad \theta(\mathsf{b}) = (1,1,1), \quad \theta(\mathsf{c}) = 0; \qquad (n_+, n_-)\big(X^{(r,s)}\big) = (r,s)$$

$$\lambda \text{ dominant}: \ \theta_1 \geq \theta_2 \geq \theta_3 \geq 0, \qquad n_+ \geq n_- \geq 0$$

$$C^Y_{\mathrm{HW}}(\lambda) = \big\{\text{colour singlets, weight } \lambda,\ Y \text{ letters}\big\} \cap \ker E_{12} \cap \ker E_{23} \cap \ker J_+, \qquad J_+ = u\,\partial_v \ \text{ on } \ X^{(r,s)} \leftrightarrow u^r v^s$$

$$C^Y_{\mathrm{HW}}(\lambda) = \bigoplus_{\text{skeletons}} C^Y_{\mathrm{HW}}(\lambda; S), \qquad S = \{(\text{type}, r+s)\} \text{ multiset}$$

$$\dim C^Y_{\mathrm{HW}}(\lambda; S) = \text{Weyl-alternated colour-singlet character (exact)}$$

$$SU(3): \quad g = \mathrm{Mult}\big(t_F \otimes t_B \otimes \tau\big), \qquad t_F: \ \gamma \mapsto V, \ \beta_I \mapsto e_{I+1} \wedge e_{I+2}, \ \mathsf{b} \mapsto e_1 \wedge e_2 \wedge e_3, \ \mathsf{c} \mapsto 1 \ \text{(column wedges of } \theta)$$

$$t_B = \prod_{n_-} [pq] \prod u, \quad [pq] = u_p v_q - v_p u_q, \qquad \tau = \text{colour trace word}, \qquad (g \pm Cg)(x_k) \text{ at probes } \Rightarrow V^\pm$$

$$h = \dim C^Y_{\mathrm{HW}} - \mathrm{rk}\, Q_0|_{Y-1} - \mathrm{rk}\, Q_0|_Y, \qquad \rho: \text{ evaluation at diagonal probes}$$

(b) $SU(N \geq 4)$: $\mathfrak{gl}(3|2)$ super highest weights

$$V = \mathbb{C}^{3|2} = \langle \theta_1, \theta_2, \theta_3 \,|\, z_{\dot 1}, z_{\dot 2} \rangle, \qquad \text{letters of degree } k: \ \Pi^{k-1}\Lambda^k V, \qquad L = \Theta + n_+ + n_-$$

$$\mu \vdash L, \quad \mu_4 \leq 2 \ \ (\text{(3|2) hook})$$

$$\lambda(\mu) = \big(\mu_1, \mu_2, \mu_3;\ \#\{i \geq 4 : \mu_i \geq 1\},\ \#\{i \geq 4 : \mu_i \geq 2\}\big)$$

$$\text{raising: } E_{12},\ E_{23},\ J_+ \ (\text{even}), \quad \theta_f\,\partial_z \ (\text{odd}), \qquad \text{distinguished Borel}$$

$$K(\mu, Y) = \dim \ker\big(H^Y(R_N)_\mu \to H^Y(C_\infty)_\mu\big)$$

$$K = \dim R^Y - \mathrm{rk} \begin{pmatrix} Q: R^Y \to R^{Y+1} & 0 \\ \iota: R^Y \to C^Y_\infty & Q_0: C^{Y-1}_\infty \to C^Y_\infty \end{pmatrix} + \mathrm{rk}\, Q_0\big|_{C^{Y-1}_\infty} - \mathrm{rk}\, Q\big|_{R^{Y-1}}$$

stable range: $R^j_N = 0 \ \ (j < 2N+2)$; routes: $\ell(\mu) \leq 2$, hooks: theorems; else exact Q and stable inclusion

1.4 Frontier

G	all sectors complete	partial / targeted
$SU(2)$	$E + J_L \leq \frac{33}{2}$ [p]	detector all sectors $\leq \frac{35}{2}$; $\frac{37}{2}$–22 partial; chains at $E + J_L = \frac{39}{2}, 20$
$SU(3)$	$E + J_L \leq 15$ [p] ($\leq \frac{23}{2}$: 2610.20340)	detector ≤ 17; windows at 16, $\frac{33}{2}$, 17, $\frac{35}{2}$
$SU(4)$	$E + J_L \leq \frac{27}{2}$ [p]	14: 34/38 shapes (incl. $K = 1$)
$SU(5)$	≤ 14	$\frac{29}{2}$: 26/29; 15: 3/53
$SU(6)$	≤ 15	$\frac{31}{2}$: 1/38
$SU(7)$	≤ 15	$\frac{31}{2}$: 19/37; 16: 14/62
$SU(8)$	$\leq \frac{31}{2}$	16: 45/56; $\frac{33}{2}$: 19/45
$SU(9)$	$\leq \frac{33}{2}$	17: 41/64; $\frac{35}{2}$: 18/50
$SU(10)$	≤ 17	$\frac{35}{2}$: 33/38; 18: 39/72
$SU(11..15)$	≤ 18	–

1.5 Fortuitous generators and classes

G	class	λ	Y	E	(R_1, R_2, R_3)	J_L	J_R	data
$SU(2)$	$\mathcal{O}_0$	$(4,4,4;0,0)$	7	$\frac{19}{2}$	$(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{5}{2}$	0	primary; 2209.06728
	$\mathcal{O}_0$ multiplet	$E + J_L \in [\frac{25}{2}, \frac{33}{2}]$	7					69 HW blocks (descendants)
	$\mathcal{O}_1$	$(6,6,6;0,0)$	9	$\frac{27}{2}$	$(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{9}{2}$	0	primary; 2304.10155
	products	$E + J_L \in [16, \frac{33}{2}]$	9					graviton $\times \mathcal{O}_0$, descendants: fresh 0
	P39	$(5,5,5;3,0)$	9	15	$(2,2,2)$	$\frac{9}{2}$	$\frac{3}{2}$	$h = 13$, known 12, fresh 1 [p,p$'$]
	X40	$(7,5,5;1,1)$	10	$\frac{31}{2}$	$(\frac{7}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{9}{2}$	0	$(h, \rho, g) = (8,8,7)$ [p,p$'$]
$SU(3)$	X_7	$(2,2,2;2,2)$	7	$\frac{21}{2}$	$(\frac{5}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{3}{2}$	0	$C = -1$, $(h, \rho, g) = (26, 26, 25)$; 35 HW blocks
	X_7 member	$(0,0,0;5,5)$	7	$\frac{27}{2}$	$(\frac{7}{2}, \frac{7}{2}, \frac{7}{2})$	$\frac{3}{2}$	0	fortuitous [$\mathbb{Q}$]; 26 trace generators
	F_1 tower	$(n{+}4,4,4;0,0)$	$n{+}7$	$n{+}\frac{19}{2}$	$(n{+}\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{5}{2}$	0	$n \geq 3$; 2412.08695, 2610.20340
	B_1 tower	$(n{+}5,5,4;0,0)$	$n{+}8$	$n{+}11$	$(n{+}2, 2, 1)$	3	0	2412.08695, 2610.20340

G	class	λ	Y	E	(R_1,R_2,R_3)	J_L	J_R	data
	new primary	$(5,4,4;1,1)$	10	$\frac{27}{2}$	$(\frac{7}{2},\frac{5}{2},\frac{5}{2})$	$\frac{5}{2}$	0	$C=-1$, $h=2$, $\rho=0$ [cond: free]
	new primary	$(7,4,4;1,0)$	11	14	$(5,2,2)$	$\frac{5}{2}$	$\frac{1}{2}$	$C=+1$, $h=2$ (F_1 desc.+1), $\rho=0$
	new primary	$(6,4,4;1,1)$	11	$\frac{29}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{5}{2}$	0	$C=+1$, $h=3$, $\rho=0$
	new primary	$(7,5,4;1,0)$	11	$\frac{29}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{3}{2})$	3	$\frac{1}{2}$	$C=+1$, $h=4$, $\rho=0$
$SU(4)$	first class	$(4,2,2;2,2)$	9	$\frac{25}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{3}{2}$	0	$C=-1$, $K=1$ [$\mathbb{Q}$]; first $SU(4)$ class
	(Codex)	$(2,2,2;3,3)$	9	$\frac{27}{2}$	$(\frac{7}{2},\frac{7}{2},\frac{7}{2})$	$\frac{3}{2}$	0	17500-term stable primitive; $=(4,2,2;2,2)-2e_1+e_++e_-$
$SU(5..15)$	–							none through the frontier

descendant shifts at fixed Y : $(-e_I;+e_\pm)\ (\theta\,\partial_z)$, $(+e_I;0)\ (\partial_\theta)$, $(7,4,4;1,0)Y11\to(6,4,4;1,1)Y11:\ (-e_1;+e_-)$

1.6 Quantum lift

class	result	level
$\mathcal{O}_0$ ($SU(2)$)	closed completion $\sum_{r\le4}\hbar^r\mathcal{O}_r$; no incoming $\Rightarrow$ all orders	[$\mathbb{Q}$]
$\mathcal{O}_1$ ($SU(2)$)	$d_1\neq0$ (2606.27955)	[p,p′]
P39	$d_1=0$; $R_2,R_3,R_4\in\operatorname{im}Q_0$; $r=5$ target $h=0$; no incoming $\Rightarrow$ all orders	[p,p′]
X40	$d_1=0$; $R_2\in\operatorname{im}Q_0+Q_1H$; $R_3,R_4\in\operatorname{im}Q_0$; $[R_5]=0$; incoming $h=0\Rightarrow$ all orders	[p,p′] ($r\le2$), [p]
X_7	$d_1=0$ in/out; d_2 out target $h=0$, in source $h=0$; $r\ge3$ empty $\Rightarrow$ all orders	[p]
F_1, $n=3$	closed completion $\mathcal{O}_0..\mathcal{O}_4$; $r\ge5$ empty; no incoming $\Rightarrow$ all orders ($d_1=0$: 2610.20340)	[p]
$(5,4,4;1,1)Y10$	$d_1=d_3=0$, $d_4=0$ [cond], no incoming; $d_2=0$ on both classes (target $(3,2,2;3,3)Y9$, $h=331$, $\rho=329$) $\Rightarrow$ all orders	[p], [cond]
$(7,4,4;1,0)Y11$	$d_1=0$ (target $h=0$), $d_2=0$ [cond] ($h=\rho=1464$), $d_3=0$, $d_4=0$ [cond], no incoming $\Rightarrow$ all orders	[p], [cond]
$(6,4,4;1,1)Y11$	no incoming; $d_3=0$; $d_4=0$ [cond]; d_1, d_2 open	[p]
$(7,5,4;1,0)Y11$	no incoming; $d_4=0$ [cond] ($h=\rho=501$); d_1,d_2,d_3 open	[p]
B_1, $(8,5,4;0,0)Y11$	d_1 target $(7,4,3;1,1)Y11$: $h=2$, $\rho=0$	open
$SU(4)$, $(4,2,2;2,2)Y9$	d_1: R_1 exact at $(3,1,1;3,3)Y9^-$ ($h=111$); R_2 exact ($h=0$); $r\ge3$ empty; incoming d_1 rank 0 (453 sources), d_2 source $h=0\Rightarrow$ all orders	[p]
(Codex) $(2,2,2;3,3)Y9$	five Q_1 scalar queries $=0$	partial

Chain data: P39, $(5,5,5;3,0)Y9$

r	target	target data	verdict
1	$(4,4,4;4,1)$, 9	$h=13$	$d_1=0$
2	$(3,3,3;5,2)$, 8	$g_8=583$	$R_2\in\operatorname{im}Q_0$ (45,450,714 terms)
3	$(2,2,2;6,3)$, 7	$h=1$	$R_3\in\operatorname{im}Q_0$ (17,787,235)
4	$(1,1,1;7,4)$, 6	$h=\rho=g=152$	$R_4\in\operatorname{im}Q_0$ (1,276,152)
5	$(0,0,0;8,5)$, 5	$h=0$	–

r	X40 target	data	verdict
1	$(6,4,4;2,2)$, 10	–	$d_1=0$ (4,429,603)
2	$(5,3,3;3,3)$, 9	–	$R_2\in\operatorname{im}Q_0+Q_1H$
3	$(4,2,2;4,4)$, 8	$g_8=1112$	$R_3\in\operatorname{im}Q_0$
4	$(3,1,1;5,5)$, 7	–	$R_4\in\operatorname{im}Q_0$
5	$(2,0,0;6,6)$, 6	$h=\rho=g=30$	$\operatorname{res}R_5=0$

Chain data: F_1, $n=3$, $(7,4,4;0,0)Y10$, $C=-1$

r	target	dim /h/ρ	residual	$\mathcal{O}_r$ (coordinates; $\operatorname{rk}Q_0$)
1	$(6,3,3;1,1)$, 10	34430/0/0	$Q_1\mathcal{O}_0$	$Y9$ (28311; 18003)
2	$(5,2,2;2,2)$, 9	63983/609/608	$Q_2\mathcal{O}_0+Q_1\mathcal{O}_1$	$Y8$ (49328; 32144)
3	$(4,1,1;3,3)$, 8	26400/0/0	$Q_2\mathcal{O}_1+Q_1\mathcal{O}_2$	$Y7$ (18919; 13063)
4	$(3,0,0;4,4)$, 7	1896/34/34	$Q_3\mathcal{O}_1+Q_2\mathcal{O}_2+Q_1\mathcal{O}_3$	$Y6$ (1169; 825)

r	$SU(4)$ $(4,2,2;2,2)Y9$: target, $C=-1$	data
1	$(3,1,1;3,3)$, 9	18133/12574/5448, $h=111$
2	$(2,0,0;4,4)$, 8	983/666/317, $h=0$
in 1	$(5,3,3;1,1)$, 9	$h=453$, $\mathrm{rk}\, d_1=0$
in 2	$(6,4,4;0,0)$, 10	$h=0$

Structural

$$\mathrm{res}(\hbar)\,\mathbf{Q}=\mathbf{Q}_{\mathrm{IR}}\,\mathrm{res}(\hbar),\quad \mathbf{Q}_{\mathrm{IR}}=0\ \Rightarrow\ \mathrm{res}(R_r)=0\ \Rightarrow\ d_r=0 \text{ at targets with } h=\rho\quad([\mathrm{cond}];\ r=1:\ 2610.22332)$$

$$SU(2),\ Y=11,\ E+J_L\in[\tfrac{41}{2},\tfrac{45}{2}]:\ \ 7 \text{ blocks},\ \dim E_2=17,\ E_2\subset \mathrm{span}(E_2\cdot E_2)$$

$$\rho-g=1:\ \mathrm{X40},\ X_7\quad(\text{Coulomb-visible fortuitous})$$

$$0\neq[X_7]\in H^7\big(\mathfrak{sl}_3[z_+,z_-],\mathfrak{sl}_3;\mathbb{C}\big)_{(5,5)},\qquad X_7(z)=4838400\quad[\mathbb{Q}]$$

1.7 Data

$SU(2)$ per level states $=\sum h\cdot\dim(\theta)_{SU(3)}\cdot(n_+-n_-+1)$

$E{+}J_L$	sect.	HW	B	F	NG blk	NG	$E{+}J_L$	sect.	HW	B	F	NG blk	NG
1	1	0	0	0	0	0	9	39	112	1786	1173	0	0
$\frac{3}{2}$	1	0	0	0	0	0	$\frac{19}{2}$	26	140	1542	2934	0	0
2	2	1	6	0	0	0	10	46	211	3327	3507	0	0
$\frac{5}{2}$	1	1	0	6	0	0	$\frac{21}{2}$	39	271	6618	3734	0	0
3	5	2	1	8	0	0	11	59	413	6807	8733	0	0
$\frac{7}{2}$	2	2	18	0	0	0	$\frac{23}{2}$	46	531	9582	13824	0	0
4	6	4	18	12	0	0	12	81	811	21483	13593	1	1
$\frac{9}{2}$	5	4	2	38	0	0	$\frac{25}{2}$	59	1053	26406	25614	1	1
5	9	6	36	30	0	0	13	94	1563	30504	46380	3	3
$\frac{11}{2}$	6	7	102	18	0	0	$\frac{27}{2}$	81	2028	63444	49640	4	5
6	16	15	59	139	0	0	14	116	2986	90111	74934	7	9
$\frac{13}{2}$	9	17	96	228	0	0	$\frac{29}{2}$	94	3835	101250	138786	7	11
7	19	29	423	114	0	0	15	150	5567	177657	170608	12	19
$\frac{15}{2}$	16	37	402	420	0	0	$\frac{31}{2}$	116	7203	280314	222792	9	19
8	26	60	345	912	0	0	16	172	10330	333117	391821	14	27
$\frac{17}{2}$	19	72	1230	714	0	0	$\frac{33}{2}$	150	13377	494548	548270	15	30

NG = odd-Y (non-graviton) highest-weight classes: all in the $\mathcal{O}_0$ multiplet at $Y=7$; $Y=9$ blocks at $E+J_L=16,\frac{33}{2}$: products/descendants (fresh 0).

$SU(N\geq 4)$ relation scan

$$\text{shape } \mu \text{ at level } 2(E+J_L)=2(\mu_1+\mu_2+\mu_3)+3\textstyle\sum_{i\geq 4}\mu_i$$

K one prime; $K(4,2,2,2,2)=1$ $[\mathbb{Q}]$, all other completed shapes: $K=0$

$SU(4)$, $\mu=(4,2,2,2,2)$, Y	5	6	7	8	9	10	11	12
$\dim R^Y$	17	213	1201	3430	5321	4579	2085	404
$\mathrm{rk}\, Q\vert_{R^Y}$	17	196	1005	2424	2894	1681	401	0
$\dim H^Y(R_4)$	0	0	0	1	3	4	3	3
K	0	0	0	0	0	**1**	0	0

$$K=4579-7900+6216-2894=1\quad[\mathbb{Q}:\ 359/467/783 \text{ primes} > \text{Hadamard}],\qquad SU(5),\ \mu=(4,2,2,2,2):\ K=0$$

2. $\mathcal{N}=4$ Yang–Mills, $G=SO(2N+1),\ Sp(N)$

2.1 Action and components

$$C^A(\theta)=ig\,\mathsf{c}^A+\theta_I\delta^{AB}(\gamma^I)_B-\tfrac{1}{2}\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A+\frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B,\qquad Q_+^I C^A=\frac{\partial_L C^A}{\partial\theta_I}$$

$$A=1,\dots,\dim G=N(2N+1)$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\int d^3\theta\Big(\tfrac{1}{2}\delta_{AB}C^A\bar\partial C^B+\tfrac{i}{3!}C_{ABC}C^AC^BC^C\Big)\Big]_{(0,2)},\qquad Q_0C^A=\tfrac{i}{2}C^A{}_{BC}C^BC^C$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+\tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$
$$(t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\qquad[t_B,t_C]=iC^A{}_{BC}t_A,\qquad\delta_{AB}=2\,\mathrm{tr}(t_At_B)$$
$$\mathsf{w}^A_{\dot a}\mapsto\tfrac{1}{2}\partial_{+\dot a}\mathsf{c}^A,\qquad-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}=-\tfrac{1}{16}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^A)$$
$$C_{ABC}\text{ of }\mathfrak{so}(2N+1)\text{ or }\mathfrak{sp}(N)$$
$$\text{components: }\mathsf{c}^A,\ \mathsf{b}_A\text{ (vector)},\quad(\gamma^I)_A,\ (\beta_I)^A\ (I=1,2,3\text{ adjoint chirals})$$

2.2 Lie algebra realization

$$SO(2n+1):\ N_F=2n+1,\quad A^{\mathsf T}\eta+\eta A=0,\quad\eta_{i\bar\imath}=1;\qquad Sp(n):\ N_F=2n,\quad A^{\mathsf T}\Omega+\Omega A=0,\quad\Omega_{i\bar\imath}=s_i$$
$$\bar\imath=N_F+1-i,\qquad s_i=+1\ (i\le n),\quad s_i=-1\ (i>n),\qquad\mathfrak{h}=\text{diagonal}$$
$$\text{basis: independent subset of }\ E_{ij}-E_{\bar\jmath\bar\imath}\ (SO),\qquad E_{ij}-s_is_jE_{\bar\jmath\bar\imath}\ (Sp)$$
$$\kappa_{RS}=\mathrm{tr}_F(t_Rt_S),\qquad\sum_R(t_R)_{ab}(t^R)_{cd}=\tfrac{1}{2}\delta_{ad}\delta_{bc}-\tfrac{1}{2}F_{ac}F_{bd},\quad F=\eta\text{ or }\Omega,\qquad C_F=\tfrac{N_F-1}{2}\ (SO),\ \tfrac{N_F+1}{2}\ (Sp)$$

trace words: $O(\infty)$ and $Sp(\infty)$ stable algebras coincide (reversal identity); C_∞ realized on $SO(n)$, $n\ge\#\text{letters}$

2.3 Highest-weight algorithm

$$G_{\mathrm{HW}},\ \lambda\text{ dominant},\ C^Y_{\mathrm{HW}}(\lambda)\text{ as 1.3(a)}$$
$$\dim C^Y_{\mathrm{HW}}:\ \text{Weyl integration, torus points summed over signed-permutation orbits (exact)}$$
$$\dim C^Y_\infty:\ \text{Burnside over }\prod S_2\wr S_k\text{ on perfect matchings}$$
$$f_Y=h_Y-\mathrm{rk}\,\pi^*,\qquad\mathrm{rk}\,\pi^*=\mathrm{rk}\begin{pmatrix}\Pi & B_G\\ Q_0^\infty & 0\end{pmatrix}-\mathrm{rk}\,Q_0^\infty-\mathrm{rk}\,B_G$$
$$\text{stable range: }Y<N_F+1\ (SO),\quad Y<n+1\ (Sp(n))\ \Rightarrow\ f=0$$

2.4 Frontier

(one prime, all sectors)

G	complete $E+J_L\le$	partial (done/total sectors)
SO(3)	12	13: 16/94, 14: 11/116
Sp(1)	12	13: 16/94, 14: 11/116
SO(5)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
Sp(2)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
SO(7)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
Sp(3)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116
SO(9)	$\frac{23}{2}$	12: 17/81, 13: 16/94, 14: 11/116
Sp(4)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116
SO(11)	12	$\frac{25}{2}$: 17/59, 13: 48/94, $\frac{27}{2}$: 1/81, 14: 25/116
Sp(5)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116

$SO(3)=Sp(1)$: $SU(2)$ census ($\le\frac{33}{2}$).

2.5 Fortuitous generators and classes

G	class	λ	Y	E	(R_1,R_2,R_3)	J_L	J_R	data
$SO(3)=Sp(1)$	$\mathcal{O}_0$	$(4,4,4;0,0)$	7	$\frac{19}{2}$	$(\frac{3}{2},\frac{3}{2},\frac{3}{2})$	$\frac{5}{2}$	0	first level
$SO(5)=Sp(2)$	first class	$(3,2,2;1,1)$	7	9	$(3,2,2)$	1	0	first level; $f=1$, $\rho=0$
$SO(7)$	O_f	$(3,3,3;0,0)$	8	$\frac{17}{2}$	$(\frac{5}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	0	first level; 2506.13887; $(h,\rho)=(1,1)$
$Sp(3)$	first class	$(5,2,2;1,1)$	9	11	$(5,2,2)$	1	0	first level; $h=f=1$
$Sp(3)$	ω	$(5,0,0;3,3)$	9	13	$(7,2,2)$	1	0	$h=1$, $\rho=0$; $\lambda=(5,2,2;1,1)-2e_2-2e_3+2e_++2e_-$
$SO(9)$	first class	$(5,3,3;0,0)$	10	$\frac{21}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	0	first level; $h=6=5+1$
$SO(11)$	class	$(7,3,3;0,0)$	12	$\frac{25}{2}$	$(\frac{13}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	0	$h=23$, $f=1$ ($E+J_L=\frac{25}{2}$ partial)
$Sp(4),Sp(5)$	–							none through $E+J_L\le12$

$SO(2N+1),\ N=3,4,5:\quad \lambda=(2N-3,3,3;0,0),\quad Y=2N+2,\quad E=\frac{4N+5}{2},\quad J_L=\frac{1}{2}$

hits above the first level (2.7): shifts $(-e_I;+e_\pm)$, $(+e_I;0)$ at fixed Y of the first class

2.6 Quantum lift

class	result	level
O_f $(SO(7))$	d_1 rank 1 into $O_m=(2,2,2;1,1)Y8$ $(h=2,\rho=1)$; 2511.09519	[p] (three primes)
$SO(7)$ $(4,3,3;0,0)Y8$	d_1 rank 1 into $(3,2,2;1,1)Y8$ $(h=11,\ \rho=10)$	[p]
$SO(7)$ $(3,1,1;2,2)Y8$	$d_1\neq 0$ into $(2,0,0;3,3)Y8$ $(h=2,\rho=1)$; $Sp(3)$: $d_1=0$	[$\mathbb{Q}$]
$SO(5)$ $(3,2,2;1,1)Y7$	out d_1, d_2 targets $h=0$; in d_1 source $h=0$; $r\geq 3$ empty $\Rightarrow$ all orders	[p]
$SO(9)$ $(5,3,3;0,0)Y10$	d_1 rank 1 into $(4,2,2;1,1)Y10$ $(h=10,\ \rho=9)$	[p]
$SO(11)$ $(7,3,3;0,0)Y12$	d_1 rank 1 into $(6,2,2;1,1)Y12$ $(h=23,\ f=1;\ Q_1\mathcal{K}\neq 0)$	[p]
$Sp(3)$ $(5,2,2;1,1)Y9$	out $d_1=0$ $(Q_1\mathcal{K}\neq 0)$; in d_1: source $(6,3,3;0,0)Y9$ $h=0$; out $d_2=0$ (target $(3,0,0;3,3)Y8$, $h=6$); $r\geq 3$ empty $\Rightarrow$ all orders	[p]
ω	no outgoing; in d_1, d_3: sources $h=0$; in d_2: from $(7,2,2;1,1)Y10$ $(h=454)$	open
$SO(7)$	$h-\rho=1$, $f=0$ (monotone); in d_1 from $(7,2,2;1,1)Y10$: rank 0 (full recheck,	[p]
$\beta=(6,1,1;2,2)Y10$	$Q_1\mathcal{K}\neq 0$; $Sp(3)$: 0); in d_2 source $h=0$	

after one loop: $(h,\rho)^{SO(2N+1)}=(h,\rho)^{Sp(N)}$ in all four lifted sectors

S-duality test of $E_2=E_\infty$

$$E_2=E_\infty\ \wedge\ \text{S-duality}\ \Longrightarrow\ \sum_{(\lambda,Y)\mapsto(E,R_I,J_L,J_R)}\dim E_2^{SO(2N+1)}=\sum_{(\lambda,Y)\mapsto(E,R_I,J_L,J_R)}\dim E_2^{Sp(N)}\quad\forall\,(E,R_I,J_L,J_R)$$

mismatch at any charge $\Rightarrow$ $d_{r\geq 2}\neq 0$ in $SO(2N+1)$ or $Sp(N)$

per level: # charges (E,R_I,J_L,J_R) with $h^{SO}\neq h^{Sp}$ and with $E_2^{SO}\neq E_2^{Sp}$, all sectors [p]

$E+J_L$	$\sum h^{SO}$	$\sum h^{Sp}$	$\#\{h^{SO}\neq h^{Sp}\}$	$\#\{d_1\neq 0\}$	$\sum\dim E_2$	$\#\{E_2^{SO}\neq E_2^{Sp}\}$	d_2 cand.	d_3 cand.	$\#\{E_4^{SO}\neq E_4^{Sp}\}$
$SO(7)$ vs $Sp(3)$									
5	12	12	0	0/0	12	0	–	–	0
$\frac{11}{2}$	14	14	0	0/0	14	0	–	–	0
6	34	34	0	0/0	34	0	–	–	0
$\frac{13}{2}$	39	39	0	0/0	39	0	–	–	0
7	79	79	0	0/0	79	0	–	–	0
$\frac{15}{2}$	100	100	0	0/0	100	0	–	–	0
8	203	203	0	0/0	203	0	–	–	0
$\frac{17}{2}$	255	255	0	0/0	255	0	–	–	0
9	497	495	**2**	1/0	495	**0**	–	–	0
$\frac{19}{2}$	649	647	**2**	1/0	647	**0**	–	–	0
10	1236	1230	**6**	3/0	1230	**0**	–	2/2 (rk ≤ 0)	0
$\frac{21}{2}$	1646	1636	**7**	4/0	1636	**0**	–	2/2 (rk ≤ 0)	0
11	3047	3029	**12**	7/0	3029	**0**	–	5/5 (rk ≤ 0)	0
$SO(9)$ vs $Sp(4)$									
5	12	12	0	0/0	12	0	–	–	0
$\frac{11}{2}$	14	14	0	0/0	14	0	–	–	0
6	34	34	0	0/0	34	0	–	–	0
$\frac{13}{2}$	39	39	0	0/0	39	0	–	–	0
7	79	79	0	0/0	79	0	–	–	0
$\frac{15}{2}$	100	100	0	0/0	100	0	–	–	0
8	207	207	0	0/0	207	0	–	–	0
$\frac{17}{2}$	261	261	0	0/0	261	0	–	–	0
9	516	516	0	0/0	516	0	–	–	0
$\frac{19}{2}$	675	675	0	0/0	675	0	–	–	0

$\#\{d_1\neq 0\}$: SO/Sp; d_r cand. : #pairs SO/Sp (max rk); $E_\infty^{SO(7)}=E_\infty^{Sp(3)}$ at every charge, $E+J_L\leq 11$

2.7 Data

h (computed sectors) / **f** per level

* stable-range sectors not counted in h; P partial level

$E{+}J_L$	$SO(3)$	$Sp(1)$	$SO(5)$	$Sp(2)$	$SO(7)$	$Sp(3)$	$SO(9)$	$Sp(4)$	$SO(11)$	$Sp(5)$
8	60	60	164	164	43^*	203	0^*	207	0^*	207
$\frac{17}{2}$	72	72	207	207	117^*	255	0^*	261	0^*	261
9	112	112	381	381	$385\,/\,\mathbf{1}^*$	495	0^*	516	0^*	516
$\frac{19}{2}$	140	140	495	495	$596\,/\,\mathbf{1}^*$	647	0^*	675	0^*	675
10	211	211	$877\,/\,\mathbf{1}$	$877\,/\,\mathbf{1}$	$1218\,/\,\mathbf{3}^*$	1230	168^*	1304	0^*	1309
$\frac{21}{2}$	271	271	$1151\,/\,\mathbf{2}$	$1151\,/\,\mathbf{2}$	$1641\,/\,\mathbf{5}^*$	1636	554^*	1736	0^*	1745
11	413	413	$1996\,/\,\mathbf{6}$	$1996\,/\,\mathbf{6}$	$3047\,/\,\mathbf{9}$	3029	$2058\,/\,\mathbf{1}^*$	3277	0^*	3308
$\frac{23}{2}$	531	531	$2665\,/\,\mathbf{9}$	$2665\,/\,\mathbf{9}$	$4099\,/\,\mathbf{11}$	4077	$3627\,/\,\mathbf{2}^*$	4426	0^*	4470
12	$811\,/\,\mathbf{1}$	$811\,/\,\mathbf{1}$	$90\,/\,\mathbf{1}^{\mathrm{p}}$	$90\,/\,\mathbf{1}^{\mathrm{p}}$	100^{p}	$7408\,/\,\mathbf{1}$	0^{p}	8252	651^*	8374
$\frac{25}{2}$	–	–	–	–	–	1385^{p}	–	1748^{p}	1642^{p}	1821^{p}
13	261^{p}	261^{p}	$726\,/\,\mathbf{7}^{\mathrm{p}}$	$726\,/\,\mathbf{7}^{\mathrm{p}}$	868^{p}	1380^{p}	842^{p}	1659^{p}	$848\,/\,\mathbf{1}^{\mathrm{p}}$	1741^{p}
$\frac{27}{2}$	–	–	–	–	–	24^{p}	–	44^{p}	50^{p}	50^{p}
14	160^{p}	160^{p}	$378\,/\,\mathbf{6}^{\mathrm{p}}$	$378\,/\,\mathbf{6}^{\mathrm{p}}$	429^{p}	424^{p}	434^{p}	442^{p}	13^{p}	447^{p}

(h,ρ) of lifted pairs

sector	$SO(2N+1)$	$Sp(N)$
$(3,3,3;0,0)Y8$	$(1,1)$	$(0,0)$
$(2,2,2;1,1)Y8$	$(2,1)$	$(1,1)$
$(4,3,3;0,0)Y8$	$(7,7)$	$(6,6)$
$(3,2,2;1,1)Y8$	$(11,10)$	$(10,10)$
$(5,3,3;0,0)Y10$	$(6,6)$	$(5,5)$
$(4,2,2;1,1)Y10$	$(10,9)$	$(9,9)$

ω orbit	$SO(7)$ h/ρ	$Sp(3)$ h/ρ
$(5,0,0;3,3)Y9$	$0/0$	$1/0$
$(6,1,1;2,2)Y10$	$355/354$	$354/354$
$(7,2,2;1,1)Y11$	$1/0$	$1/0$
$(8,3,3;0,0)Y12$	$41/40$	$41/40$

3. $\mathcal{N}=2$ $\mathbb{Z}_2$ quiver, $G=SU(N)\times SU(N)$

3.1 Action and components

$$V=V_0\oplus V_1\subset\mathfrak{gl}(2N),\qquad V_0=\left\{\begin{pmatrix}a&0\\0&b\end{pmatrix}:\operatorname{tr}a=\operatorname{tr}b=0\right\}=\mathfrak{su}(N)\oplus\mathfrak{su}(N)$$

$$V_1=\left\{\begin{pmatrix}0&x\\y&0\end{pmatrix}\right\}=(\mathbf{N},\bar{\mathbf{N}})\oplus(\bar{\mathbf{N}},\mathbf{N})$$

$$t_A \text{ Hermitian basis of } V,\quad \operatorname{tr}(t_At_B)=\tfrac12\delta_{AB},\qquad C_{ABC}=-2i\operatorname{tr}\big(t_A[t_B,t_C]\big),\qquad \sigma=\operatorname{diag}(\mathbf{1}_N,-\mathbf{1}_N)$$

$$C^A(\theta)=ig\,\mathsf{c}^A+\theta_I\delta^{AB}(\gamma^I)_B-\tfrac12\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A+\frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B,\qquad Q^I_+C^A=\frac{\partial_LC^A}{\partial\theta_I}$$

$$A=1,\dots,\dim V$$

$$\sigma\,C(\theta_1,\theta_2,\theta_3)\,\sigma^{-1}=C(-\theta_1,-\theta_2,\theta_3)\;\Rightarrow\;\mathsf{c},\mathsf{b},\gamma^3,\beta_3\in V_0,\qquad \gamma^1,\gamma^2,\beta_1,\beta_2\in V_1$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\int d^3\theta\Big(\tfrac12\delta_{AB}C^A\bar\partial C^B+\tfrac{i}{3!}C_{ABC}C^AC^BC^C\Big)\Big]_{(0,2)},\qquad Q_0C^A=\tfrac{i}{2}C^A{}_{BC}C^BC^C$$

$$g_1=g_2=g$$

$$\begin{aligned}S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[&\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\big|_{A,B,C\in V_0}\\&+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B\\&+iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}\end{aligned}$$

$$A\in V_0;\qquad B,C\in V_0\ (I=3),\qquad B,C\in V_1\ (I=1,2);\qquad D,E\in V_1,\ F\in V_0$$

$$(t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\qquad [t_B,t_C]=iC^A{}_{BC}t_A,\qquad \delta_{AB}=2\operatorname{tr}(t_At_B)$$

$$\mathsf{w}^A_{\dot a}\mapsto\tfrac12\partial_{+\dot a}\mathsf{c}^A,\qquad -\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}=-\tfrac{1}{16}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^A)$$

components: $\underbrace{\mathsf{c},\mathsf{b}}_{\text{two vector multiplets}}$, $\underbrace{\gamma^3,\beta_3}_{\text{two adjoint chirals}}$

$$\underbrace{\gamma^1,\gamma^2,\beta_1,\beta_2}_{\text{two bifundamental hypermultiplets}}$$

$\kappa=g_2/g_1$: $Q_0\ni\kappa^2$ on the node-2 block of the hypermultiplet moment maps

$$Q_n \text{ weights: } \prod_{\text{inputs}} w\prod_{\text{vertices}}\text{coef}\prod_{\text{lines}}\text{prop}$$

$$\#\kappa^2 \text{ in a term of } Q_n = \#(\mathsf{b}_2, \beta_{3,2} \text{ consumed}) - \#(\mathsf{b}_2, \beta_{3,2} \text{ produced})$$
$$\Rightarrow\ Q_n(\kappa) = \phi_\kappa Q_n(1) \phi_\kappa^{-1}\ \ \forall n\ \Rightarrow\ E_r(\kappa) \cong E_r(1)\ \ (\kappa \neq 0)$$

3.2 Lie algebra realization

$$N_F = 2N, \qquad \mathfrak{g} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : \operatorname{tr} a = \operatorname{tr} b = 0 \right\}, \qquad \text{bifundamental lines: } E_{ij}\ (i \leq N < j \text{ or } j \leq N < i),\ \text{dual } E_{ji}$$
$$\text{node-2 generators: } t^R \to \kappa^2 t^R; \qquad \text{weights: } U(N) \times U(N)/U(1)_{\text{diag}} \text{ torus } \Rightarrow \text{ baryon-number-zero invariants}$$

3.3 Highest-weight algorithm

$$G_{\text{HW}} = GL(2)_{12} \times GL(1)_3 \times SL(2)_{\dot{a}}, \qquad \theta_1 \geq \theta_2 \geq 0,\ \ \theta_3 \geq 0,\ \ n_+ \geq n_- \geq 0$$
$$C^Y_{\text{HW}}(\lambda) = \{\text{colour singlets}\} \cap \ker E_{12} \cap \ker J_+, \qquad \text{columns: } \theta_1 \text{ wedge columns of } (\theta_1, \theta_2)\ \oplus\ \theta_3 \text{ single slots}$$

3.4 Frontier

(all sectors)

$$SU(2)^2 :\ E + J_L \leq \tfrac{21}{2}; \qquad SU(3)^2 :\ E + J_L \leq 9$$

3.5 Classes and quantum lift

Classes f: open (no stable comparison); E_2 classes per level in 3.6

$$E_2 = h - \operatorname{rk} d_1^{\text{out}} - \operatorname{rk} d_1^{\text{in}}; \qquad d_2 \text{ candidates: } E_2^Y(\lambda) > 0,\ E_2^{Y-1}(\lambda - 2\mathsf{d}) > 0$$
$$\operatorname{rk} d_2 = 0 \ \text{ on every candidate pair, } \ \kappa^2 = 1, 9$$
$$SU(2)^2 :\ 37 \text{ pairs } (Q_2\mathcal{K} \neq 0 :\ 29), \qquad SU(3)^2 :\ 26 \text{ pairs } (Q_2\mathcal{K} \neq 0 :\ 9), \qquad \max \operatorname{rk} d_2 = 0$$
$$d_3 :\ (3,3,3;0,0)Y6 \to (0,0,0;3,3)Y4 :\ \operatorname{rk} = 0\ \ (Q_3\mathcal{K} = 0)$$
$$\text{sources } (E + J_L \leq 9) :\ \rho = 0$$
$$\text{targets } (h, \rho) :\ (1,1,0;2,2)Y4\ (6,4),\ (2,0,0;2,2)Y4\ (2,0),\ (1,1,1;2,2)Y4\ (20,14),\ (2,0,1;2,2)Y4\ (6,0)$$
$$\text{control: } Q_2 \to 4Q_2\ \Rightarrow\ \operatorname{rk} d_2 = 3$$

3.6 Data

Columns

$$\text{HW blocks} = \#\{(\lambda, Y) : h > 0\}, \qquad \textstyle\sum h = \sum_{(\lambda,Y)} h$$
$$\#\{d_1 \neq 0\} = \#\{(\lambda, Y) : \operatorname{rk} d_1 > 0\}, \qquad \sum E_2 = \sum h - 2 \sum \operatorname{rk} d_1$$
$$d_2 \text{ pairs} = \#\left\{(\lambda, Y) : E_2^Y(\lambda) > 0,\ E_2^{Y-1}(\lambda - 2\mathsf{d}) > 0\right\}\ \ (\text{where } d_2 \text{ can act})$$
$$\max \operatorname{rk} d_2 = 0\ \Rightarrow\ \text{no two-loop lift}$$
$$\#\{Q_2\mathcal{K} \neq 0\} = \#\{\text{pairs with } Q_2 \neq 0 \text{ on the } E_2 \text{ cocycles}\}\ \ (\text{non-trivial tests})$$

N	$E{+}J_L$	sectors	HW blocks	$\sum h$	$\#\{d_1 \neq 0\}$	$\sum E_2$	d_2 pairs	$\max \operatorname{rk} d_2$	$\#\{Q_2\mathcal{K} \neq 0\}$ $(\kappa^2 = 1, 9)$
2	2	4	3	4	0	4	0	–	–
2	$\frac{5}{2}$	2	1	2	0	2	0	–	–
2	3	8	4	6	0	6	0	–	–
2	$\frac{7}{2}$	4	4	6	0	6	0	–	–
2	4	13	11	17	0	17	0	–	–
2	$\frac{9}{2}$	8	7	14	0	14	0	–	–
2	5	20	19	38	1	36	0	–	–
2	$\frac{11}{2}$	13	15	42	1	40	0	–	–
2	6	31	34	93	2	89	0	–	–
2	$\frac{13}{2}$	20	26	116	1	112	0	–	–
2	7	44	57	240	3	230	1	0	0
2	$\frac{15}{2}$	31	43	276	3	268	1	0	0
2	8	61	90	560	5	544	3	0	4
2	$\frac{17}{2}$	44	73	716	4	696	1	0	2
2	9	84	140	1298	7	1258	7	0	10
2	$\frac{19}{2}$	61	110	1690	8	1640	5	0	8
2	10	111	205	3153	13	3059	11	0	18
2	$\frac{21}{2}$	84	169	4088	13	3960	8	0	16
3	2	4	3	4	0	4	0	–	–

N	$E{+}J_L$	sectors	HW blocks	$\sum h$	$\#\{d_1\neq 0\}$	$\sum E_2$	d_2 pairs	$\max \mathrm{rk}\, d_2$	$\#\{Q_2\mathcal{K}\neq 0\}$ $(\kappa^2=1,9)$
3	$\frac{5}{2}$	2	1	2	0	2	0	–	–
3	3	8	5	8	0	8	0	–	–
3	$\frac{7}{2}$	4	5	8	0	8	0	–	–
3	4	13	12	21	0	21	0	–	–
3	$\frac{9}{2}$	8	8	18	0	18	0	–	–
3	5	20	24	54	0	54	0	–	–
3	$\frac{11}{2}$	13	19	62	0	62	0	–	–
3	6	31	40	149	1	147	1	0	0
3	$\frac{13}{2}$	20	32	176	1	174	0	–	–
3	7	44	69	400	4	390	2	0	0
3	$\frac{15}{2}$	31	54	478	3	468	1	0	0
3	8	61	113	1016	8	986	8	0	4
3	$\frac{17}{2}$	44	92	1304	7	1270	3	0	2
3	9	84	171	2634	19	2542	11	0	12

4. $\mathcal{N}=2$ SCQCD, $SU(N)$, $N_f=2N$

4.1 Action and components

$$V_0^{(1)}=\Big\{\begin{pmatrix}a&0\\0&0\end{pmatrix}:\operatorname{tr}a=0\Big\}=\mathfrak{su}(N),\qquad V_1=\Big\{\begin{pmatrix}0&x\\y&0\end{pmatrix}\Big\},\quad x,y\in\mathbb{C}^{N\times N}$$

$$t_A \text{ Hermitian basis of } V_0^{(1)}\oplus V_1,\quad \operatorname{tr}(t_At_B)=\tfrac12\delta_{AB},\qquad C_{ABC}=-2i\operatorname{tr}\big(t_A[t_B,t_C]\big)$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\big|_{A,B,C\in V_0^{(1)}}$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$$A\in V_0^{(1)};\qquad B,C\in V_0^{(1)}\ (I=3),\qquad B,C\in V_1\ (I=1,2);\qquad D,E\in V_1,\ F\in V_0^{(1)}$$

$$(t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\qquad \mathsf{w}^A_{\dot a}\mapsto\tfrac12\partial_{+\dot a}\mathsf{c}^A$$

components: $\underbrace{\mathsf{c}^A,\mathsf{b}_A}_{\text{vector}}$, $\underbrace{(\gamma^3)_A,(\beta_3)^A}_{\text{adjoint chiral}}$, $\underbrace{(\gamma^{1,2})_B,(\beta_{1,2})^B,\ B\in V_1}_{2N\text{ fundamental hypermultiplets}}$; $SU(N)_2\subset U(2N)_{\text{flavour}}$ singlets

$=\kappa\to 0$ limit of the $\mathbb{Z}_2$ quiver (node 2 vector multiplet decoupled)

4.2 Lie algebra realization and highest weights

$$\mathfrak{g}=\Big\{\begin{pmatrix}a&0\\0&0\end{pmatrix}:\operatorname{tr}a=0\Big\},\qquad \text{node-2 roots: Haar measure only;}\qquad G_{\mathrm{HW}}\text{ as 3.3}$$

4.3 Frontier

(all sectors)

$$SU(2),\ N_f=4:\ E+J_L\le 12;\qquad SU(3),\ N_f=6:\ E+J_L\le 9$$

4.4 Classes and quantum lift

Classes f: open (no stable comparison); E_2 classes per level in 4.5

$$SU(2):\ \#\{d_1\neq 0\}=0\ \ (E+J_L\le 12)$$
$$\mathrm{rk}\,d_2=0 \text{ on every candidate pair},\qquad \max\mathrm{rk}\,d_2=0$$
$$SU(2):\ 42\text{ pairs }(Q_2\mathcal{K}\neq 0:\ 41),\qquad SU(3):\ 13\text{ pairs }(Q_2\mathcal{K}\neq 0:\ 5)$$
$$SU(3):\ d_1\neq 0:\ (1,1,5;0,0)Y6,\ (1,1,4;1,0)Y6,\ (1,1,3;1,1)Y6,\ (1,1,6;0,0)Y6$$
$$(1,1,5;1,0)Y6,\ (1,1,4;1,1)Y6,\ (1,1,4;2,0)Y6,\ (1,1,7;0,0)Y8$$

4.5 Data

Columns

$$\text{HW blocks} = \#\{(\lambda, Y) : h > 0\}, \qquad \sum h = \sum_{(\lambda,Y)} h$$

$$\#\{d_1 \neq 0\} = \#\{(\lambda, Y) : \operatorname{rk} d_1 > 0\}, \qquad \sum E_2 = \sum h - 2 \sum \operatorname{rk} d_1$$

$$d_2 \text{ pairs} = \#\big\{(\lambda, Y) : E_2^Y(\lambda) > 0,\ E_2^{Y-1}(\lambda - 2\mathsf{d}) > 0\big\} \ \text{(where } d_2 \text{ can act)}$$

$$\max \operatorname{rk} d_2 = 0 \ \Rightarrow\ \text{no two-loop lift}$$

$$\#\{Q_2\mathcal{K} \neq 0\} = \#\{\text{pairs with } Q_2 \neq 0 \text{ on the } E_2 \text{ cocycles}\} \ \text{(non-trivial tests)}$$

N	$E{+}J_L$	sectors	HW blocks	$\sum h$	$\#\{d_1 \neq 0\}$	$\sum E_2$	d_2 pairs	$\max \operatorname{rk} d_2$	$\#\{Q_2\mathcal{K} \neq 0\}$
2	2	4	3	3	0	3	0	–	–
2	$\frac{5}{2}$	2	1	1	0	1	0	–	–
2	3	8	3	4	0	4	0	–	–
2	$\frac{7}{2}$	4	3	4	0	4	0	–	–
2	4	13	7	10	0	10	0	–	–
2	$\frac{9}{2}$	8	4	6	0	6	0	–	–
2	5	20	9	21	0	21	0	–	–
2	$\frac{11}{2}$	13	9	21	0	21	0	–	–
2	6	31	19	37	0	37	0	–	–
2	$\frac{13}{2}$	20	14	54	0	54	0	–	–
2	7	44	26	112	0	112	0	–	–
2	$\frac{15}{2}$	31	23	109	0	109	0	–	–
2	8	61	42	232	0	232	1	0	0
2	$\frac{17}{2}$	44	30	307	0	307	1	0	1
2	9	84	59	481	0	481	1	0	1
2	$\frac{19}{2}$	61	54	623	0	623	1	0	1
2	10	111	87	1263	0	1263	4	0	4
2	$\frac{21}{2}$	84	75	1527	0	1527	5	0	5
2	11	144	122	2632	0	2632	8	0	8
2	$\frac{23}{2}$	111	106	3850	0	3850	7	0	7
2	12	186	155	6440	0	6440	14	0	14
3	2	4	3	3	0	3	0	–	–
3	$\frac{5}{2}$	2	1	1	0	1	0	–	–
3	3	8	4	5	0	5	0	–	–
3	$\frac{7}{2}$	4	4	5	0	5	0	–	–
3	4	13	9	12	0	12	0	–	–
3	$\frac{9}{2}$	8	6	8	0	8	0	–	–
3	5	20	14	25	0	25	0	–	–
3	$\frac{11}{2}$	13	12	26	0	26	0	–	–
3	6	31	26	55	0	55	0	–	–
3	$\frac{13}{2}$	20	23	66	0	66	0	–	–
3	7	44	40	150	1	148	1	0	0
3	$\frac{15}{2}$	31	35	161	1	159	1	0	0
3	8	61	69	323	2	319	2	0	0
3	$\frac{17}{2}$	44	56	424	1	420	2	0	1
3	9	84	101	781	3	773	7	0	4

5. Massive-adjoint flows

5.1 Action and components

$\mathcal{N} = 4$ $SU(N)$, one mass $\quad N = 2, 3;\ A = 1, \dots, N^2 - 1;\ [t_B, t_C] = iC^A{}_{BC} t_A,\ \operatorname{tr}(t_A t_B) = \frac{1}{2}\delta_{AB},\ (t_A^{\mathrm{adj}})^C{}_B = iC^C{}_{AB}$

$$S_{\mathrm{tw},E} = \int_{\mathbb{C}^2} \Omega \wedge \Big[\mathsf{b}_A \big(\bar\partial \mathsf{c}^A - \tfrac{g}{2} C^A{}_{BC} \mathsf{c}^B \mathsf{c}^C \big)$$

$$+ \sum_{I=1}^{3} \big(\bar\partial (\gamma^I)_B - ig\, \mathsf{c}^A (t_A^{\mathrm{adj}})^C{}_B (\gamma^I)_C \big) (\beta_I)^B$$

$$+ \tfrac{i}{3!} \varepsilon_{IJK} C_{ABC} \delta^{AD} \delta^{BE} \delta^{CF} (\gamma^I)_D (\gamma^J)_E (\gamma^K)_F + \tfrac{M}{2}\, \delta^{AB} (\gamma^3)_A (\gamma^3)_B - \tfrac{1}{4} \mathsf{w}^A_{\dot a} \mathsf{w}^{A\dot a} \Big]_{(0,2)}$$

SCQCD $SU(N)$ + mass $\quad N=2,3$; index ranges, t_A, C_{ABC} as 4.1

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\big|_{A,B,C\in V_0^{(1)}}$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F+\tfrac{M}{2}\,\delta^{AB}(\gamma^3)_A(\gamma^3)_B-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$\mathbb{Z}_2$ quiver $SU(2)\times SU(2)$ + masses index ranges, t_A, C_{ABC} as 3.1; $\Lambda^{AB}=M\delta^{AB}$ ($A,B\in$ node 1), $-M\delta^{AB}$ ($A,B\in$ node 2)

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\big|_{A,B,C\in V_0}$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F+\tfrac{1}{2}\Lambda^{AB}(\gamma^3)_A(\gamma^3)_B-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$$Q_0'=Q_0+\Lambda^{AB}(\gamma^3)_B\frac{\partial}{\partial(\beta_3)^A},\qquad Q_n'=Q_n\ (n\geq 1),\qquad \Lambda^{AB}=M\delta^{AB}\ \text{(one node)}$$

UV	mass	IR
$\mathcal{N}=4$ $SU(2)$, $SU(3)$	M on γ^3	Leigh–Strassler; $\bar W_{\mathrm{IR}}=-\frac{1}{2M}\,\mathrm{Tr}[\gamma^1,\gamma^2]^2$
SCQCD $SU(2)$, $SU(3)$	M on γ^3	$\mathcal{N}=1$ SQCD, $N_f=2N$, quartic $\bar W$
$\mathbb{Z}_2$ quiver $SU(2)^2$	$(M_1,M_2)=(M,-M)$	Klebanov–Witten

5.2 Gradings and charges

$$(A,B;n_+,n_-;\mathrm{gh}):\quad A=\theta_1+\theta_3,\quad B=\theta_2+\theta_3,\quad \mathrm{gh}=Y-\Theta;\qquad C(A,B;n_\pm;\mathrm{gh})=\bigoplus_{\theta_3}C^Y(\lambda)$$
$$R_1+R_3=A+\mathrm{gh},\quad R_2+R_3=B+\mathrm{gh},\quad J_L=\tfrac{1}{2}(n_++n_--\mathrm{gh}),\quad J_R=\tfrac{1}{2}(n_+-n_-)$$
$$R_{\mathrm{IR}}=\tfrac{1}{2}(R_1+R_2)+R_3,\qquad E_{\mathrm{IR}}=\tfrac{3}{2}R_{\mathrm{IR}}+2J_L,\qquad w=A+B+2(n_++n_-)=2R_{\mathrm{IR}}+4J_L=\tfrac{4}{3}\big(E_{\mathrm{IR}}+J_L\big)$$
$$d_r:\ (A,B;n_\pm;\mathrm{gh})\to(A-2r,B-2r;n_\pm+r;\mathrm{gh}+1+2r),\qquad w\ \text{fixed}$$
$$G_{\mathrm{HW}}=GL(2)_{12}\times SL(2)_{\dot a}:\quad A\geq B\geq 0,\ \ n_+\geq n_-\geq 0;\qquad \text{Lie algebra as 1.2, 3.2, 4.2}$$

5.3 Frontier

theory	complete	d_r pairs ($r=2,3$)	$\max\mathrm{rk}\,d_r$
$\mathcal{N}$=4 $SU(2)$, one mass	$w\leq 16$ (partial: $w=17$)	101	0
$\mathcal{N}$=4 $SU(3)$, one mass	$w\leq 13$	49	0
SCQCD $SU(2)$ + mass	$w\leq 15$	27	0
SCQCD $SU(3)$ + mass	$w\leq 13$	13	0
$\mathbb{Z}_2$ quiver $SU(2)^2$, masses $(M,-M)$	$w\leq 13$	16	0

5.4 Classes and quantum lift

theory	class	$(A,B;n_+,n_-;\mathrm{gh})$	$(R_1{+}R_3,R_2{+}R_3)$	J_L	R_{IR}	E_{IR}	quantum
$SU(2)$, one mass	w	$(3,3;0,0;-2)$	$(1,1)$	1	1	$\frac{7}{2}$	graviton
	w^2	$(6,6;0,0;-4)$	$(2,2)$	2	2	7	closed completion to $\hbar^3$ (quartic $\bar W$)
	$\mathcal{O}_7$	$(5,2;0,0;-2)$	$(3,0)$	1	$\frac{3}{2}$	$\frac{17}{4}$	$d_1=0$; $r\geq 2$ empty $\Rightarrow$ all orders [p]
	$\mathcal{O}_{16}$	$(8,8;0,0;-5)$	$(3,3)$	$\frac{5}{2}$	3	$\frac{19}{2}$	all orders [p]

$$\mathcal{O}_7,\mathcal{O}_9,\mathcal{O}_{14},\mathcal{O}_{16}:\ 2512.12674$$
$$\mathcal{O}_{16}:\ d_1\ (h_T=0),\ d_2\ (h_T=7,\ Q_2\mathcal{K}\neq 0),\ d_3\ (h_T=8),\ d_4\ (Q_4\mathcal{K}=0)=0;\ \text{no } d_5,\ \text{no incoming}$$
$$SU(2),\ w\leq 17:\ d_1\neq 0\ \text{only at}\ (6,6;1,1;-3);\qquad \mathrm{rk}\,d_2=\mathrm{rk}\,d_3=0\ \text{on every candidate pair}$$
$$\mathbb{Z}_2\ \text{quiver}:\ d_1\neq 0\ \text{at}\ (2,2;k,1;1),\ k=1,2,3$$
$$\text{other theories: } d_1=0,\qquad \mathrm{rk}\,d_2=\mathrm{rk}\,d_3=0$$

Massive SQED analogue

theory	$(7,7;1,0;-5)$	$(7,7;1,0;-4)$	$(7,7;1,0;-6)$
$U(1)$, $\mathcal{I} = ig\mathsf{c}(\gamma_+\beta^+ - \gamma_-\beta^-) + \frac{m}{2}\gamma_s^2 + y\gamma_s\gamma_+\gamma_-$	$d_2 \neq 0$ (closed form)		
$\mathcal{N}=4$ $SU(2)$, one mass	$h = 0$	$h = 1$, $d_2 = 0$ ($Q_2\mathcal{K} \neq 0$)	$h = 0$
$\mathcal{N}=4$ $SU(3)$, one mass	$h = 0$, d_2 rank 0	running	
SCQCD $SU(2)$ + mass	$E_1 = 0$		

5.5 Data

Columns as 3.6 with $(\lambda, Y) \to (A, B; n_\pm; \mathrm{gh})$; d_r pairs: $E_2 > 0$ at source and target of d_r, $r = 2, 3$; all $\mathrm{rk}\, d_r = 0$

theory	w	sectors	HW blocks	$\sum h$	$\#\{d_1 \neq 0\}$	$\sum E_2$	d_r pairs	r tested
$\mathcal{N}$=4 $SU(2)$, one mass	4	7	4	4	0	4	–	–
	5	7	4	4	0	4	–	–
	6	13	11	11	0	11	–	–
	7	13	12	13	0	13	–	–
	8	22	25	30	0	30	–	–
	9	22	24	40	0	40	–	–
	10	34	51	81	0	81	–	–
	11	34	49	115	0	115	–	–
	12	50	93	218	0	218	4	d_2
	13	50	90	317	0	317	4	d_2
	14	70	154	569	0	569	11	d_2
	15	70	151	845	0	845	14	d_2, d_3
	16	95	240	1473	1	1471	32	d_2, d_3
	17	95	241	2194	0	2194	36	d_2, d_3
$\mathcal{N}$=4 $SU(3)$, one mass	4	7	6	6	0	6	–	–
	5	7	10	10	0	10	–	–
	6	13	19	23	0	23	–	–
	7	13	27	42	0	42	–	–
	8	22	50	96	0	96	1	d_2
	9	22	58	171	0	171	1	d_2
	10	34	98	358	0	358	5	d_2
	11	34	112	641	0	641	7	d_2
	12	50	178	1284	0	1284	15	d_2
	13	50	194	2300	0	2300	20	d_2, d_3
SCQCD $SU(2)$ + mass	4	7	8	11	0	11	–	–
	5	7	0	0	0	0	–	–
	6	13	16	46	0	46	–	–
	7	13	0	0	0	0	–	–
	8	22	34	194	0	194	–	–
	9	22	0	0	0	0	–	–
	10	34	62	826	0	826	2	d_2
	11	34	0	0	0	0	–	–
	12	50	109	3371	0	3371	7	d_2
	13	50	0	0	0	0	–	–
	14	70	173	13622	0	13622	18	d_2, d_3
	15	70	0	0	0	0	–	–
SCQCD $SU(3)$ + mass	4	7	8	11	0	11	–	–
	5	7	0	0	0	0	–	–
	6	13	18	44	0	44	–	–
	7	13	0	0	0	0	–	–
	8	22	35	203	0	203	1	d_2
	9	22	0	0	0	0	–	–
	10	34	71	934	0	934	3	d_2
	11	34	0	0	0	0	–	–
	12	50	125	4430	0	4430	9	d_2
	13	50	0	0	0	0	–	–
$\mathbb{Z}_2$ quiver $SU(2)^2$, masses $(M, -M)$	4	7	10	16	0	16	–	–
	5	7	0	0	0	0	–	–
	6	13	26	66	0	66	–	–
	7	13	0	0	0	0	–	–
	8	22	53	299	1	297	–	–
	9	22	0	0	0	0	–	–
	10	34	98	1306	1	1304	3	d_2
	11	34	0	0	0	0	–	–
	12	50	170	5442	1	5440	13	d_2, d_3
	13	50	0	0	0	0	–	–

6. $\mathcal{N} = 4$ $SU(3)$: β and Leigh–Strassler deformations

6.1 Action and components

$$A=1,\dots,8,\qquad [t_B,t_C]=iC^A{}_{BC}t_A,\quad \mathrm{tr}(t_At_B)=\tfrac12\delta_{AB},\quad (t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\quad d_{ABC}=2\,\mathrm{tr}\big(t_A\{t_B,t_C\}\big)$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)$$
$$+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B$$
$$+\tfrac{1+q}{4}\cdot\tfrac{i}{3!}\,\delta^{AD}\delta^{BE}\delta^{CF}\big(\varepsilon_{IJK}C_{ABC}-iS_{IJK}d_{ABC}\big)(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F-\tfrac14\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$$S_{IJK}=\tfrac{1-q}{1+q}|\varepsilon_{IJK}|+\tfrac{2h}{1+q}\delta_{IJK},\qquad \delta_{IJK}=1\ (I=J=K),\ 0\ \text{else}$$

$$\tfrac{1+q}{4}\cdot\tfrac{i}{3!}\,\delta^{AD}\delta^{BE}\delta^{CF}\big(\varepsilon_{IJK}C_{ABC}-iS_{IJK}d_{ABC}\big)(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F$$
$$=\mathrm{tr}\,\gamma^1\gamma^2\gamma^3-q\,\mathrm{tr}\,\gamma^1\gamma^3\gamma^2+\tfrac{h}{3}\sum_I\mathrm{tr}(\gamma^I)^3,\qquad \gamma^I=(\gamma^I)_At_A$$

$$\underbrace{q=1,\ h=0}_{\mathcal{N}=4\ (\times\frac12)},\qquad \underbrace{h=0}_{\beta},\qquad \underbrace{(q,h)\ \text{generic}}_{\text{Leigh–Strassler}};\qquad SU(2):\ d_{ABC}=0\ \Rightarrow\ \text{deformation trivial}$$

components: c^A, b_A (vector), $(\gamma^I)_A$, $(\beta_I)^A$ ($I=1,2,3$ adjoint chirals)

$$Q_0'\beta_I=Q_0\beta_I\big|_{\varepsilon C\to\varepsilon C-iSd}$$

Q_n' : fan rule, $\gamma\gamma\gamma$ vertex at slot k : $\varepsilon_{IJK}\hat A_n\,[L,R]\ +\ S_{IJK}\hat A_n\,\{L,R\}_{\mathfrak{su}(3)}$

6.2 Lie algebra realization

$$\mathfrak{su}(3)\ \text{as 1.2};\qquad \mathrm{tr}(t_At_B)=\tfrac12\delta_{AB},\qquad \{t_A,t_B\}=\tfrac13\delta_{AB}\mathbf{1}+d_{ABC}t_C$$
$$\mathrm{tr}(t_At_Bt_C)=\tfrac14\big(d_{ABC}+iC_{ABC}\big),\qquad \{L,R\}_{\mathfrak{su}(3)}=\{L,R\}-\tfrac23\,\mathrm{tr}(LR)\mathbf{1}$$

6.3 Highest-weight algorithm

$$\beta:\ G_{\mathrm{HW}}=U(1)^3_\theta\times SL(2)_{\dot a},\qquad C^Y(\theta_1,\theta_2,\theta_3;n_+,n_-)=\{\text{colour singlets}\}\cap\ker J_+\ \ \text{(no flavour raising)}$$
$$\mathrm{LS}:\ U(1)^3\to U(1)\times\mathbb{Z}_3,\qquad \mathcal{C}^Y(\Theta,c;n_+,n_-)=\bigoplus_{\theta_1+\theta_2+\theta_3=\Theta,\ \theta_2+2\theta_3\equiv c\ (3)}C^Y(\theta;n_+,n_-)$$

$$Q_n'=Q_n^{(0)}+\sum_I Q_n^{(h,I)},\qquad Q_n^{(h,I)}:\ \theta\mapsto\theta-n(1,1,1)+3e_I-(1,1,1)$$

$$d_r:\ (\Theta,c;n_\pm)_Y\to(\Theta-3r,c;n_\pm+r)_{Y+1-r}$$

conserved: $E,\ J_L,\ J_R,\ R_1+R_2+R_3=\tfrac12(3Y-\Theta),\ c$ (LS); cyclic $\gamma^1\to\gamma^2\to\gamma^3$: $c\mapsto c+\Theta$

6.4 Frontier

$$\beta,\ \mathrm{LS}:\ E+J_L\le 9\ \ \text{(all sectors, through } d_3)$$

6.5 Fortuitous classes

theory	class	$(\Theta,c;n_+,n_-)_Y$	E	$\sum R_I$	J_L	J_R	data
LS	fortuitous	$(7,0;0,0)_5$	6	4	1	0	$h^{SU(3)}=5,\ h^{SU(4,5,6)}=0$
LS	fortuitous	$(7,0;1,0)_5$	7	4	$\frac32$	$\frac12$	$h^{SU(3)}=5,\ h^{N\to\infty}=0$

$SU(N\ge Y+1)$: no trace relations at $Y\ \Rightarrow\ h_Y^{N\to\infty}$; full fortuity classification: open

6.6 Quantum lift

$d_2 \neq 0$, LS $SU(3)$ [p]; seeds, p', $(q,h) = (-\frac{3889}{4341}, \frac{976546}{421077}), (4, \frac{85}{6}), (1, \frac{4321}{97})$: $\mathrm{rk}\, d_2 = 1$; $h = 0$: $\mathrm{rk}\, d_2 = 0$

source	target	$(E, J_L, J_R) \to (E, J_L, J_R)$	$Q_2\mathcal{K}$	$\mathrm{rk}\, d_2$
$(7,0;0,0)_5$	$(1,0;2,2)_4$	$(6,1,0) \to (\frac{13}{2}, \frac{1}{2}, 0)$	0	1
$(8,0;0,0)_6$	$(2,0;2,2)_5$	$(7,1,0) \to (\frac{15}{2}, \frac{1}{2}, 0)$	0	1
$(7,0;1,0)_5$	$(1,0;3,2)_4$	$(7, \frac{3}{2}, \frac{1}{2}) \to (\frac{15}{2}, 1, \frac{1}{2})$	$\neq 0$	1

$$Q_2\mathcal{K} = 0: \quad d_2[\mathcal{O}_0] = [Q_1\mathcal{O}_1], \qquad Q_0'\mathcal{O}_1 = -Q_1'\mathcal{O}_0$$

$$(7,0;0,0)_5: \; h_{E_1} = 5 \xrightarrow{d_1,\ \mathrm{rk}\,3} h_{E_2} = 2 \xrightarrow{d_2,\ \mathrm{rk}\,1} h_{E_3} = h_{E_\infty} = 1; \qquad \Theta \not\equiv 0\ (3): \ \times 3 \text{ cyclic images}$$

β: $\mathrm{rk}\, d_2 = 0$ on all 11 candidate pairs; β, LS: no d_3 candidates $(E + J_L \leq 9)$

Gates $(Q_0')^2 = 0$, $\{Q_0', Q_1'\} = 0$, $\sum_j Q_j Q_{2-j} = 0$, $\sum_j Q_j Q_{3-j} = 0$ (nonzero terms; β and LS); controls: d-slot sign, no $\mathfrak{su}(3)$ projection, Q_2/Q_3 d-part $\times(-1)$, h-vertex $\times 2$: violations

6.7 Data

β

j	sectors	$\sum h_{E_1}$	d_1 pairs	$d_1 \neq 0$	d_2 pairs	$d_2 \neq 0$	d_3 pairs	$d_3 \neq 0$	$\sum h_{E_4}$
2	1	0	0	0	0	0	0	0	0
3	1	0	0	0	0	0	0	0	0
4	2	2	0	0	0	0	0	0	2
5	1	1	0	0	0	0	0	0	1
6	5	5	1	0	0	0	0	0	5
7	2	3	0	0	0	0	0	0	3
8	6	6	0	0	0	0	0	0	6
9	5	9	1	0	0	0	0	0	9
10	9	14	2	0	0	0	0	0	14
11	6	16	1	0	0	0	0	0	16
12	16	44	5	0	1	0	0	0	44
13	9	40	3	0	0	0	0	0	40
14	19	71	8	2	0	0	0	0	67
15	16	108	8	1	1	0	0	0	106
16	26	149	17	1	4	0	0	0	147
17	19	190	13	4	1	0	0	0	182
18	39	383	25	7	4	0	0	0	369

j	source	target	E_2	E_2	$Q_2\mathcal{K} \neq 0$	$\mathrm{rk}\, d_2$	$(E, J_L, J_R) \to (E, J_L, J_R)$
12	$(2,2,2;0,0)_5$	$(0,0,0;2,2)_4$	2	1	0	0	$(\frac{11}{2}, \frac{1}{2}, 0) \to (6,0,0)$
15	$(2,2,2;1,0)_5$	$(0,0,0;3,2)_4$	6	1	0	0	$(\frac{13}{2}, 1, \frac{1}{2}) \to (7, \frac{1}{2}, \frac{1}{2})$
16	$(3,3,2;0,0)_6$	$(1,1,0;2,2)_5$	2	1	0	0	$(7,1,0) \to (\frac{15}{2}, \frac{1}{2}, 0)$
16	$(4,2,2;0,0)_5$	$(2,0,0;2,2)_4$	1	2	4	0	$(\frac{13}{2}, \frac{3}{2}, 0) \to (7,1,0)$
16	$(4,2,2;0,0)_6$	$(2,0,0;2,2)_5$	1	2	0	0	$(7,1,0) \to (\frac{15}{2}, \frac{1}{2}, 0)$
16	$(4,2,2;0,0)_7$	$(2,0,0;2,2)_6$	1	1	0	0	$(\frac{15}{2}, \frac{1}{2}, 0) \to (8,0,0)$
17	$(3,2,2;1,0)_6$	$(1,0,0;3,2)_5$	9	1	0	0	$(\frac{15}{2}, 1, \frac{1}{2}) \to (8, \frac{1}{2}, \frac{1}{2})$
18	$(2,2,2;2,0)_5$	$(0,0,0;4,2)_4$	9	2	4330	0	$(\frac{15}{2}, \frac{3}{2}, 1) \to (8,1,1)$
18	$(3,3,3;0,0)_6$	$(1,1,1;2,2)_5$	2	2	2672	0	$(\frac{15}{2}, \frac{3}{2}, 0) \to (8,1,0)$
18	$(3,3,3;0,0)_7$	$(1,1,1;2,2)_6$	1	5	0	0	$(8,1,0) \to (\frac{17}{2}, \frac{1}{2}, 0)$
18	$(5,2,2;0,0)_7$	$(3,0,0;2,2)_6$	3	2	0	0	$(8,1,0) \to (\frac{17}{2}, \frac{1}{2}, 0)$

LS

j	classes	$\sum h_{E_1}$	d_1 pairs	$d_1 \neq 0$	d_2 pairs	$d_2 \neq 0$	d_3 pairs	$d_3 \neq 0$	$\sum h_{E_4}$
2	1	0	0	0	0	0	0	0	0
3	2	0	0	0	0	0	0	0	0
4	1	2	0	0	0	0	0	0	2
5	1	1	0	0	0	0	0	0	1
6	6	3	0	0	0	0	0	0	3
7	1	3	0	0	0	0	0	0	3
8	3	6	0	0	0	0	0	0	6
9	6	11	1	0	0	0	0	0	11
10	3	9	0	0	0	0	0	0	9
11	3	14	1	0	0	0	0	0	14
12	12	56	3	1	0	0	0	0	54
13	3	26	2	0	0	0	0	0	26
14	6	54	3	1	1	1	0	0	46
15	12	136	6	3	1	0	0	0	118
16	6	88	6	1	1	1	0	0	84
17	6	143	5	2	2	1	0	0	121
18	20	460	14	9	3	0	0	0	382

j	source	target	E_2	E_2	$Q_2\mathcal{K}\neq 0$	$\mathrm{rk}\, d_2$	$(E,J_L,J_R)\to(E,J_L,J_R)$
14	$(7,0;0,0)_5$	$(1,0;2,2)_4$	2	2	0	1	$(6,1,0)\to(\frac{13}{2},\frac{1}{2},0)$
15	$(6,0;1,0)_5$	$(0,0;3,2)_4$	2	1	0	0	$(\frac{13}{2},1,\frac{1}{2})\to(7,\frac{1}{2},\frac{1}{2})$
16	$(8,0;0,0)_6$	$(2,0;2,2)_5$	3	3	0	1	$(7,1,0)\to(\frac{15}{2},\frac{1}{2},0)$
17	$(7,0;1,0)_5$	$(1,0;3,2)_4$	2	4	2112	1	$(7,\frac{3}{2},\frac{1}{2})\to(\frac{15}{2},1,\frac{1}{2})$
17	$(7,0;1,0)_6$	$(1,0;3,2)_5$	3	1	0	0	$(\frac{15}{2},1,\frac{1}{2})\to(8,\frac{1}{2},\frac{1}{2})$
18	$(6,0;2,0)_5$	$(0,0;4,2)_4$	4	2	5680	0	$(\frac{15}{2},\frac{3}{2},1)\to(8,1,1)$
18	$(9,0;0,0)_7$	$(3,0;2,2)_6$	8	4	0	0	$(8,1,0)\to(\frac{17}{2},\frac{1}{2},0)$
18	$(9,1;0,0)_7$	$(3,1;2,2)_6$	6	3	0	0	$(8,1,0)\to(\frac{17}{2},\frac{1}{2},0)$

N-dependence of h_{E_1} (LS)

class	Y	$SU(3)$	$SU(4)$	$SU(5)$	$SU(6)$	$SU(7)$
$(7,0;0,0)$	5	5	0	0	0	0
$(7,0;0,0)$	6	0	16	0	0	0
$(7,0;0,0)$	7	0			11	23
$(1,0;2,2)$	4	2	2		2	
$(8,0;0,0)$	6	4	11	2		2
$(2,0;2,2)$	5				8	
$(7,0;1,0)$	5	5	0	0	0	
$(7,0;1,0)$	6	10	41	9	9	
$(1,0;3,2)$	4				4	

7. $\mathcal{N}=4$ Yang–Mills, $G=G_2$

7.1 Action and components

$$C^A(\theta)=ig\,\mathsf{c}^A+\theta_I\delta^{AB}(\gamma^I)_B-\tfrac{1}{2}\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A+\frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B,\qquad Q^I_+C^A=\frac{\partial_L C^A}{\partial\theta_I}$$

$$A=1,\dots,14$$

$$S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[\int d^3\theta\Big(\tfrac{1}{2}\delta_{AB}C^A\bar\partial C^B+\tfrac{i}{3!}C_{ABC}C^AC^BC^C\Big)\Big]_{(0,2)},\qquad Q_0C^A=\tfrac{i}{2}C^A{}_{BC}C^BC^C$$

$$\begin{aligned}S_{\mathrm{tw},E}=\int_{\mathbb{C}^2}\Omega\wedge\Big[&\mathsf{b}_A\big(\bar\partial\mathsf{c}^A-\tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big)\\&+\sum_{I=1}^{3}\big(\bar\partial(\gamma^I)_B-ig\,\mathsf{c}^A(t_A^{\mathrm{adj}})^C{}_B(\gamma^I)_C\big)(\beta_I)^B\\&+\tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}\end{aligned}$$

$$(t_A^{\mathrm{adj}})^C{}_B=iC^C{}_{AB},\qquad[t_B,t_C]=iC^A{}_{BC}t_A,\qquad\delta_{AB}=2\,\mathrm{tr}_7(t_At_B)$$

$$\mathsf{w}^A_{\dot a}\mapsto\tfrac{1}{2}\partial_{+\dot a}\mathsf{c}^A,\qquad-\tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}=-\tfrac{1}{16}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^A)$$

$$C_{ABC}\text{ of }\mathfrak{g}_2\ (7.2)$$

components: c^A, b_A (vector), $(\gamma^I)_A$, $(\beta_I)^A$ ($I=1,2,3$ adjoint chirals)

7.2 Lie algebra realization

$$\mathfrak{g}_2=\big\{A\in\mathfrak{gl}(7):\ A^{\mathsf T}\eta+\eta A=0,\ \textstyle\sum_d\big(A_{da}\phi_{dbc}+A_{db}\phi_{adc}+A_{dc}\phi_{abd}\big)=0\big\},\qquad\eta_{i\bar\imath}=1,\ \ \bar\imath=8-i$$

$$\phi=e^{123}+2\,e^{567}+e^{147}+e^{246}+e^{345}\qquad(\dim\mathfrak{g}_2=14\text{ checked})$$

$$\mathfrak{h}=\{\mathrm{diag}(a_1,a_2,-a_1-a_2,0,a_1+a_2,-a_2,-a_1)\}$$

roots (a_1,a_2)-weights: short $\pm(1,0),\pm(0,1),\pm(1,1)$; long $\pm(1,-1),\pm(2,1),\pm(1,2)$; $\alpha>0\Leftrightarrow 2\alpha_1+\alpha_2>0$

$\kappa_{RS}=\mathrm{tr}_7(t_Rt_S)$, $C_F=\frac{14}{7}=2$; $\sum_R(t_R)_{ab}(t^R)_{cd}$ involves $\phi\ \Rightarrow\ Q_1,Q_2,Q_3$ as explicit sums over R

checks: $\dim S^k(\mathfrak{g}_2)^{G_2}=1,1,2,2,2,3,3$ ($k=2,4,\dots,14$) $\leftrightarrow$ Casimir degrees $2,6$

$\dim\Lambda^k(\mathfrak{g}_2)^{G_2}=1$ ($k=0,3,11,14$)

$\dim\mathrm{Hom}_{G_2}(\mathfrak{g}_2^{\otimes 4},\mathbb{C})=5$ ($\mathfrak{g}_2\otimes\mathfrak{g}_2=\mathbf{1}+\mathbf{14}+\mathbf{27}+\mathbf{77}+\mathbf{77}'$)

$Q_0Q_1+Q_1Q_0=0$ at $(3,3,3;0,0)Y5$, $(4,3,3;0,0)Y6$, $(3,3,2;1,0)Y5$

nonzero entries 275, 5254, 7646; violations 0

7.3 Highest-weight algorithm

G_{HW}, λ dominant, $C^Y_{\mathrm{HW}}(\lambda)$ as 1.3(a); $\quad \dim C^Y_{\mathrm{HW}}$: Weyl integration on the full torus grid μ_M^2 (exact)

generators: products of tr_7 of letter words

span $= \dim C^Y_{\mathrm{HW}}$ in every sector computed (no ϕ-networks needed)

$\rho = \mathrm{rk}\big(H \to$ values on $\mathfrak{h}$-valued letters$\big)$

7.4 Frontier

(one prime, all sectors)

$$E + J_L \le 10$$

7.5 Classes

$\rho = 0$ class in the scalar sector [p], two primes

$\lambda = (2,2,2;0,0)$, $Y = 6$: $\quad E = 6$, $R = (2,2,2)$, $J_L = J_R = 0$ $\quad$ (six γ's, flavour singlet)

$$\dim C^6 = \dim\big(S_{(2,2,2)}\mathfrak{g}\big)^G \qquad \text{(Cauchy: } \mathrm{Sym}(\mathfrak{g}\otimes\mathbb{C}^3) = \textstyle\bigoplus_\mu S_\mu\mathfrak{g}\otimes S_\mu\mathbb{C}^3\text{)}$$

$$= \dim(S^2\mathfrak{g}^{\otimes 3})^G - 2\dim(\mathfrak{g}\otimes S^2\mathfrak{g}\otimes S^3\mathfrak{g})^G + \dim(S^3\mathfrak{g}^{\otimes 2})^G + \dim(S^4\mathfrak{g}\otimes\mathfrak{g}^{\otimes 2})^G - \dim(S^4\mathfrak{g}\otimes S^2\mathfrak{g})^G$$

$$\dim C^5 = \dim\big(\mathfrak{g}\otimes S_{(2,1,1)}\mathfrak{g}\big)^G \qquad (\beta_I\gamma^4 :\ \Lambda^2\mathbb{C}^3\otimes S_\mu\mathbb{C}^3 \ni (2,2,2) \Leftrightarrow \mu = (2,1,1))$$

$$= \dim(\mathfrak{g}^{\otimes 3}\otimes S^2\mathfrak{g})^G - \dim(\mathfrak{g}\otimes S^2\mathfrak{g}^{\otimes 2})^G - \dim(\mathfrak{g}^{\otimes 2}\otimes S^3\mathfrak{g})^G + \dim(\mathfrak{g}\otimes S^4\mathfrak{g})^G$$

$h \ge \dim C^6 - \dim C^5$, $\quad \mathrm{rk}\,\mathfrak{h} = 2 \ \Rightarrow\ \rho = 0$ $\quad$ (det ^{2} of three vectors in $\mathfrak{h}\cong\mathbb{C}^2$)

G	$\dim C^5$	$\dim C^6$	h	ρ
$SU(3)$	4	3	0	0
$SO(5)$, $Sp(2)$	4	4	0	0
$\mathbf{G_2}$	**3**	**4**	**1**	**0**
$SO(7)$, $Sp(3)$	4	5	1	1
$SU(4)$	5	5	1	1

in: none; $\quad$ out: $d_1 = 0$, $d_{r\ge 2}$ no targets $\Rightarrow$ all orders

representative

$$H^{(2,2,2;0,0)}_{Y=6} = \mathbb{C}\left[\det_{IJ}\mathrm{tr}_7(\gamma^I\gamma^J)\right] = \mathbb{C}\left[\tfrac{1}{6}\,\varepsilon_{IJK}\varepsilon_{LMN}\,\mathrm{tr}_7(\gamma^I\gamma^L)\,\mathrm{tr}_7(\gamma^J\gamma^M)\,\mathrm{tr}_7(\gamma^K\gamma^N)\right]$$

$$C^6 = \mathrm{span}\big\{\varepsilon_{IJK}\varepsilon_{LMN}\,M(\gamma^I,\gamma^J,\gamma^K,\gamma^L,\gamma^M,\gamma^N)\big\}, \quad M = \text{products of traces, cycle types } (2,2,2),(2,4),(3,3),(6)$$

$$Q_0C^5 = \mathrm{span}\big\{\varepsilon_{IJK}\varepsilon_{LMN}\,M'([\gamma^I,\gamma^J],\gamma^K,\gamma^L,\gamma^M,\gamma^N),\ \varepsilon_{IJK}\varepsilon_{LMN}\,M'([\gamma^I,\gamma^L],\gamma^J,\gamma^K,\gamma^M,\gamma^N)\big\}$$

G	$\mathrm{rk}\,C^6$	$\mathrm{rk}\,Q_0C^5$	$\det T \in Q_0C^5$	cycle types outside Q_0C^5	$\det T\vert_{\mathfrak{h}^3}$
$SU(3)$	3	3	yes	–	0
$SO(5)$, $Sp(2)$	4	4	yes	–	0
$\mathbf{G_2}$	4	3	**no**	$(2,2,2)$	0
$SO(7)$, $Sp(3)$	5	4	no	$(2,2,2)$	$\neq 0$

$T^{IJ} = \mathrm{tr}_F(\gamma^I\gamma^J)$; $\quad$ ranks: evaluation at random points mod 1000003 (two seeds)

$\det T \in Q_0C^5$ for rank-2 classical groups (traces in F; coefficients at two primes)

$$\varepsilon\varepsilon[X] := \varepsilon_{IJK}\varepsilon_{LMN}X, \qquad [\gamma^I,\gamma^J]\varepsilon_{IJK} \propto Q_0\beta_K$$

$$SU(3):\ \det T = \varepsilon\varepsilon\Big[-\tfrac{5}{6}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^K\gamma^L\gamma^M\gamma^N) - \tfrac{2}{3}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^L\gamma^K\gamma^M\gamma^N) + \tfrac{5}{18}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^K)\operatorname{tr}(\gamma^L\gamma^M\gamma^N)\Big]$$

$$SO(5):\ \det T = \varepsilon\varepsilon\Big[\tfrac{10}{3}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^K\gamma^L\gamma^M\gamma^N) - \tfrac{4}{3}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^L\gamma^K\gamma^M\gamma^N)$$
$$-\tfrac{2}{3}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^K)\operatorname{tr}(\gamma^L\gamma^M\gamma^N) + \operatorname{tr}([\gamma^I,\gamma^J]\gamma^L\gamma^N)\operatorname{tr}(\gamma^K\gamma^M)\Big]$$

$$Sp(2):\ \det T = \varepsilon\varepsilon\Big[-\tfrac{1}{6}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^K\gamma^L\gamma^M\gamma^N) - \tfrac{5}{6}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^L\gamma^K\gamma^M\gamma^N) + \tfrac{1}{4}\operatorname{tr}([\gamma^I,\gamma^J]\gamma^L\gamma^N)\operatorname{tr}(\gamma^K\gamma^M)\Big]$$

$$G_2:\ \det T \notin \operatorname{span}\big\{\varepsilon\varepsilon[M'([\gamma^I,\gamma^J],\dots)],\ \varepsilon\varepsilon[M'([\gamma^I,\gamma^L],\dots)]\big\}$$

independent recomputation $\quad[\mathbb{Q}]$

$$\mathfrak{g}_2 = \operatorname{stab}_{\mathfrak{so}(7)}\big(e^{123}+e^{145}+e^{167}+e^{246}-e^{257}-e^{347}-e^{356}\big), \quad \delta\text{-metric}$$

$$(\dim C^5, \dim C^6) = (3,4) \quad \text{(Racah–Speiser; constant term; two character routes)}$$

$$\operatorname{rk}_{\mathbb{Q}} C^6 = 4, \quad \operatorname{rk}_{\mathbb{Q}} Q_0C^5 = 3, \quad \operatorname{rk}_{\mathbb{Q}}(Q_0C^5 + \mathbb{C}\det T) = 4$$

$$\text{ideal in the weight basis of } S_{(2,2,2)}\mathfrak{g}_2 \text{ (no traces)}:\ h = 1$$

$$\det T\big|_{\text{commuting triples}} = 0 \quad \text{(every 3-dim abelian subalgebra of } \mathfrak{g}_2 \text{ is nilpotent)}$$

$$(\det T)^2 \in Q_0C \quad (H^{(4,4,4;0,0)}_{Y=12} = 0,\ \dim C^{12} = 30)$$

origin: $\mathcal{N}=1$ chiral ring

$$\psi_1,\psi_2\in\mathfrak{g}\otimes(\text{odd}),\quad \{\psi_\alpha,\psi_\beta\}=0,\quad S=\mathrm{tr}(\psi_1\psi_2);\qquad \text{CDSW: } S^{h^\vee}=0,\quad S^{h^\vee-1}\neq 0$$
$$X_I=\psi_1 u_I+\psi_2 v_I\quad (u_I,v_I\ \text{odd}):\quad [X_I,X_J]=0,\qquad \mathrm{tr}(X_IX_J)=-S\,(u_Iv_J+u_Jv_I)$$
$$\det_{I,J\le d}\mathrm{tr}(X_IX_J)=c_d\,S^d\prod_I u_Iv_I,\quad c_d\neq 0\quad\Rightarrow\quad \big(S^d\neq 0\ \Rightarrow\ \det T\notin I\big)$$
$$\det T\neq 0\ \text{in}\ \mathbb{C}[\mathcal{C}_d(\mathfrak{g})]^G\ \text{and}\ \det T|_{\mathfrak{h}^d}=0\quad\Longleftarrow\quad \mathrm{rk}\,\mathfrak{g}<d<h^\vee(\mathfrak{g})$$

$\mathfrak{g}$	$(\mathrm{rk},h^\vee)$	$d=3$	$d=4$	window $\mathrm{rk}<d<h^\vee$
$\mathfrak{su}(3)$	$(2,3)$	$\det T\in I$	–	$\varnothing$
$\mathfrak{so}(5)=\mathfrak{sp}(2)$	$(2,3)$	$\in I$	$\in I$	$\varnothing$
$\mathfrak{g}_2$	$(2,4)$	$\notin I,\ \rho=0$	$\in I$	$\{3\}$
$\mathfrak{su}(4)$	$(3,4)$	$\notin I,\ \rho\neq 0$	$\in I$	$\varnothing$
$\mathfrak{sp}(3)$	$(3,4)$	$\notin I,\ \rho\neq 0$	$\in I$	$\varnothing$
$\mathfrak{so}(7)$	$(3,5)$	$\notin I,\ \rho\neq 0$	$\notin I,\ \rho=0$	$\{4\}$

random evaluation, full spanning sets $\varepsilon\varepsilon M([x,x],\dots)$ (both placements)

$\mathfrak{so}(7),\ d=4:\ \mathrm{rk}\,Q_0C=8,\quad \mathrm{rk}(Q_0C+\mathbb{C}\det T)=9=\dim C^8$ (three primes)

status in the literature

statement	scope
$\mathbb{C}[\mathcal{C}_d(\mathfrak{g})]^G\to\mathbb{C}[\mathfrak{h}^d]^W$ isomorphism (Chen–Ngô, Conj. 3.1; Conj. 3.2 $\Rightarrow$ injective)	conjectured: all reductive G, all d (1905.04741, 2102.01849)
proved	GL_n (Vaccarino; Domokos), Sp_{2n} (Chen–Ngô 2102.01849), O_n, SO_n (Song–Xia–Xu 2207.03147, Thm 1.2), all d; all G, $d=2$ (Li–Nadler–Yun 2301.02618 conditional; Sempliner–Yang 2609.27360; Goyal 2610.05018 for $\mathfrak{g}_2$)
reduced version $\mathbb{C}[\mathcal{C}_d(\mathfrak{g})_{\mathrm{red}}]^G\cong\mathbb{C}[\mathfrak{h}^d]^W$	types A–D, G_2 (Hunziker 1997)
$\mathfrak{g}_2$, $d=3$: $\det T\neq 0$, $\det T\|_{\mathfrak{h}^3}=0$	contradicts Conj. 3.1, 3.2; not found in the literature
$\mathfrak{so}(7)$, $d=4$: $\det T\neq 0$, $\det T\|_{\mathfrak{h}^4}=0$	contradicts Song–Xia–Xu Thm 1.2
$S^{h^\vee-1}\neq 0$	CDSW; Etingof–Kac math/0305175, Etingof math/0312452, Kumar, Cederwall–Ferretti 2311.12119
$SO(7)$ class at $(2,2,2;1,1)Y8$	Chang–Lin 2510.24008, Chang 2604.23549: $\varepsilon\varepsilon$ Gram determinant with one gaugino-dressed factor ($d_{\mathrm{eff}}=4$)
$\mathcal{N}=4$: $[S^m]\neq 0$, $\rho(S^m)=0$, $\mathrm{rk}<m<h^\vee$, $S=\mathrm{Tr}(\lambda_+\lambda_-)$	Kibe–van Leuven–Mathieson 2610.22332: eq. (1) (all $\mathfrak{g}$, via CDSW); §4.5 direct check $\mathfrak{so}(7)$–$\mathfrak{so}(12)$; §5, App. D: $\mathfrak{so}(7)$ $S^4=$ descendant of the $(2,2,2;1,1)Y8$ class, fortuitous Q_1-partner of S^4
$\mathcal{O}(\mathcal{C}_E)^G_{(2^d)}=\mathbb{S}_{(2^d)}(E)\otimes L_d$, $\dim L_d=[d<h^\vee]$, all super E: S^d ($E=\Pi k^2$), $\det T$ ($E=k^d$), $\mathcal{N}=4$ module ($E=k^{3\|2}$) share L_d	note (notes/Gram_determinants_commuting_schemes.pdf), Thm 2.3, Cor 8.1

pure-scalar blocks ($n_\pm=0$, $Y=\Theta$): h vs ρ

G	$E+J_L\le$	blocks	$h\neq\rho$
$SO(7)$, $Sp(3)$	11	24, 24	none
$SO(9)$, $Sp(4)$	$\frac{19}{2}$	14, 14	none
G_2	10	14	$(2,2,2)Y6$ only

$$G_2:\ \det T\cdot\mathrm{tr}_7(\gamma^{(I}\gamma^{J)}),\ \det T\cdot(\text{quartics})\in Q_0C$$
$$(h=\rho\ \text{at}\ (4,2,2),(3,3,2),(4,4,2),\dots,\ Y=8,10)$$

7.6 Quantum lift

$$\#\{d_1\neq 0\}=0,\quad d_2\text{ candidates: none},\quad d_3\text{ candidates: 2 }(E+J_L=10),\ \mathrm{rk}=0\qquad (E+J_L\le 10)$$

7.7 Data

$E{+}J_L$	sectors	fails	$\sum h$	$\sum(h-\rho)$	$\#\{d_1\neq 0\}$	$\sum \dim E_2$	d_2 cand.	d_3 cand.
4	6	0	5	0	0	5	–	–
$\frac{9}{2}$	5	0	5	0	0	5	–	–
5	9	0	10	0	0	10	–	–
$\frac{11}{2}$	6	0	12	0	0	12	–	–
6	16	0	27	1	0	27	–	–
$\frac{13}{2}$	9	0	31	1	0	31	–	–
7	19	0	58	3	0	58	–	–
$\frac{15}{2}$	16	0	73	5	0	73	–	–
8	26	0	137	9	0	137	–	–
$\frac{17}{2}$	19	0	171	11	0	171	–	–
9	39	0	307	20	0	307	–	–
$\frac{19}{2}$	26	0	397	23	0	397	–	–
10	46	0	693	43	0	693	–	2 (rk ≤ 0)

8. Abelian gauge theories, $G = U(1)^k$

8.1 Action and components

$\boxed{U(1)^k \text{ with chirals } \gamma_g \text{ of charges } q_g^a}$ $a = 1,\dots,k$; g runs over the chirals of the table; $\sum_g q_g^a = 0$ on every monomial of $\mathcal{W}$

$$S_{\mathrm{tw},E} = \int_{\mathbb{C}^2} \Omega \wedge \Big[\sum_{a=1}^{k} \mathsf{b}_a\, \bar\partial \mathsf{c}^a + \sum_g \Big(\bar\partial\gamma_g - ig\sum_{a=1}^{k} q_g^a\, \mathsf{c}^a\gamma_g\Big)\beta^g + \mathcal{W}(\gamma) - \tfrac14 \sum_{a=1}^{k} \mathsf{w}_{\dot a}^a \mathsf{w}^{a\dot a}\Big]_{(0,2)}$$

$$\mathsf{w}_{\dot a}^a \mapsto \tfrac12 \partial_{+\dot a}\mathsf{c}^a, \qquad \mathcal{I} = -ig\sum_{g,a} q_g^a\, \mathsf{c}^a\gamma_g\beta^g + \mathcal{W}(\gamma) - \tfrac14\sum_a \mathsf{w}_{\dot a}^a \mathsf{w}^{a\dot a} \qquad \text{(no } \mathsf{bcc} \text{ vertex)}$$

components: $\underbrace{\mathsf{c}^a,\ \mathsf{b}_a}_{U(1)\text{ vectors}}$, $\underbrace{\gamma_g,\ \beta^g}_{\text{chirals}}$; letters $X^{(r,s)} = \partial_{+\dot1}^r \partial_{+\dot2}^s X$, c only with $r+s \geq 1$

theory	k; charges	$\mathcal{W}$	(w_g), D
$\mathcal{N}{=}2$ $U(1)$, $N_f{=}1$	1; $q_\pm = \pm1$, $q_s = 0$	$y\,\gamma_s\gamma_+\gamma_-$	$(1,1,1)$, 3
$\mathcal{N}{=}2$ $U(1)$, $N_f{=}1$ + mass	as above	$\frac{m}{2}\gamma_s^2 + y\,\gamma_s\gamma_+\gamma_-$	$(2,1,1)$, 4
$U(1)$, quartic SQED	1; $q_\pm = \pm1$	$\lambda\gamma_+^2\gamma_-^2$	$(1,1)$, 4
$U(1)$, cubic singlet	as $N_f{=}1$	$y\,\gamma_s\gamma_+\gamma_- + \frac{h}{3}\gamma_s^3$	$(1,1,1)$, 3
$U(1)$, $\gamma_s^2\gamma_+\gamma_-$	as $N_f{=}1$	$y\,\gamma_s^2\gamma_+\gamma_-$	$(1,1,1)$, 4
$\mathcal{N}{=}2$ $U(1)$, $N_f{=}2$	1; $q_{\pm}, q_{\pm'} = \pm1$, $q_s = 0$	$y\,\gamma_s(\gamma_+\gamma_- + \gamma_{+'}\gamma_{-'})$	$(1,\dots,1)$, 3
+ mass	as above	$\frac{m}{2}\gamma_s^2 + y\,\gamma_s(\gamma_+\gamma_- + \gamma_{+'}\gamma_{-'})$	$(2,1,\dots,1)$, 4
$U(1)^2$ triangle	2; $q_{x,y,z} = (1,0),(0,1),(-1,-1)$ $q_{u,v,w} = -q_{x,y,z}$	$y_1\gamma_x\gamma_y\gamma_z + y_2\gamma_u\gamma_v\gamma_w$	$(1,\dots,1)$, 3
$U(1)^2$ triangle + mass	as above, $q_s = (0,0)$	$\cdots + \frac{m}{2}\gamma_s^2 + y_3\gamma_s\gamma_u\gamma_x$	$w_z = w_w = w_s = 2$, else 1; 4
$\mathcal{N}{=}2$ $U(1)^2$ quiver	2; $q_{x,y} = \pm(1,0)$, $q_{u,v} = \pm(1,-1)$ $q_{p,r} = \pm(0,1)$, $q_s = q_t = (0,0)$	$\gamma_s(\gamma_x\gamma_y + \gamma_u\gamma_v) + \gamma_t(\gamma_p\gamma_r - \gamma_u\gamma_v)$	$(1,\dots,1)$, 3

exponent vectors of $\mathcal{W}$ independent $\Longrightarrow$ $\gamma_g \to t_g\gamma_g$, $\beta^g \to t_g^{-1}\beta^g$ sets every coupling to 1; $\qquad g = 1$

8.2 Lie algebra realization

$$\mathfrak{u}(1)^k:\ [t_a,t_b] = 0, \qquad q_g^a \in \mathbb{Z}$$

$$Q_0\mathsf{c}^a = 0, \quad Q_0\gamma_g = \sum_a q_g^a \mathsf{c}^a\gamma_g, \quad Q_0\beta^g = \partial_{\gamma_g}\mathcal{W} - \sum_a q_g^a \mathsf{c}^a\beta^g, \quad Q_0\mathsf{b}_a = -\sum_g q_g^a\beta^g\gamma_g$$

gauge charge $\sum q^a = 0$ (grading), $\qquad$ flavour lattice $= \ker_{\mathbb{Z}}\big(\text{exponent matrix of } \mathcal{W}\big)$

8.3 Blocks and charges

$$\mathrm{wt}(\gamma_g) = w_g, \quad \mathrm{wt}(\beta^g) = D - w_g, \quad \mathrm{wt}(\mathsf{b}) = D, \quad \mathrm{wt}(\mathsf{c}) = 0; \qquad \mathrm{gh}(\gamma,\beta,\mathsf{b},\mathsf{c}) = (0,-1,-2,1)$$

$$\text{block } (\mathrm{wt}; n_+, n_-; \mathrm{gh}; \text{flavour}), \qquad Q_n:\ (\mathrm{wt}; n_\pm; \mathrm{gh}) \to (\mathrm{wt} - nD;\ n_\pm + n;\ \mathrm{gh} + 2n + 1)$$

$$R = \tfrac{2}{D}\,\mathrm{wt} + \mathrm{gh}, \qquad J_L = \tfrac12(n_+ + n_- - \mathrm{gh}), \qquad J_R = \tfrac12(n_+ - n_-), \qquad (R; J_L, J_R)(\mathbf{Q}) = (1; -\tfrac12, 0)$$

$$(n_+,n_-) = (\lceil J/2\rceil, \lfloor J/2\rfloor):\ J_R \in \{0,\tfrac12\} \text{ weight space of } SL(2)_{\dot a}; \qquad \partial_{+\dot a}\text{-descendants separated}$$

8.4 Frontier

(all blocks; d_2, d_3, d_4 on every block with $\ker d_1 \neq 0$)

theory	wt	$J = n_+ + n_-$	blocks	$\ker d_1 \neq 0$	d_3 tested	d_4 tested	d_4 target $\neq 0$
$\mathcal{N}$=2 $U(1)$, N_f=1	6–20	0–4	13402	302	302	296	51
$U(1)$, cubic singlet	6–18	0–4	3869	33	26	22	3
$U(1)$, $\gamma_s^2\gamma_+\gamma_-$	8–14	0–3	1382	70	65	0	0
$\mathcal{N}$=2 $U(1)$, N_f=1 + mass	8–24	0–3	3498	20	12	12	1
$U(1)$, quartic SQED	8–18	0–3	1292	18	14	0	0
$\mathcal{N}$=2 $U(1)$, N_f=2	6–14	0–3	16944	264	264	264	9
$\mathcal{N}$=2 $U(1)$, N_f=2 + mass	8–16	0–3	4828	248	248	248	2
$U(1)$, N_f=2 quartic	8–14	0–2	1604	102	102	0	0
$U(1)$, two massive singlets	8–17	0–2	4840	33	33	33	0
$U(1)$, off-diagonal mass	8–17	0–2	9913	142	142	142	1
$U(1)^2$ triangle	6–12	0–2	13949	40	37	37	0
$U(1)^2$ triangle + mass	8–15	0–2	8463	23	18	18	0

8.5 Spectator algebra and Massey products

Spectator lemma

$$\mathcal{A} = \mathbb{C}\big[\partial^r_{+\dot 1}\partial^s_{+\dot 2}\mathsf{c}^a \ (r+s\geq 1),\ \partial^r_{+\dot 1}\partial^s_{+\dot 2}\gamma_s \ (q_s=0)\big], \qquad \underbrace{\text{no } \mathsf{b}\text{-leg}}_{\mathsf{bcc}\text{ absent}} \wedge \underbrace{\text{no neutral } \beta\text{-leg}}_{\mathcal{W}(\gamma)}$$

$$\Longrightarrow\ Q_n(aX) = (-1)^{|a|}a\,Q_n(X)\quad \forall a\in\mathcal{A},\ n\geq 0 \qquad\Longrightarrow\qquad E_r\ \mathcal{A}\text{-modules},\ d_r\ \mathcal{A}\text{-linear}$$

$$G \text{ semisimple}:\ f_{ABC}\mathsf{b}^A\mathsf{c}^B\mathsf{c}^C \text{ and } \beta\mathsf{c}\gamma \text{ contract every letter} \Longrightarrow \mathcal{A}=\mathbb{C}$$

Torsion and Massey products

$$q_x+q_y=0:\quad \partial_{+\dot 1}\mathsf{c}\cdot\gamma_x\gamma_y = \tfrac{1}{q_x}Q_0\big(\partial_{+\dot 1}\gamma_x\,\gamma_y\big)$$

$$a\,Q_1x = Q_0w \ \Longrightarrow\ d_2[a\,x] = (-1)^{|a|}\big[a\,Q_2x - Q_1w\big]$$

$$aZ = Q_0h_1,\ ZY = Q_0h_2 \ \Longrightarrow\ \langle a,Z,Y\rangle = h_1Y \pm a\,h_2 \in E_1$$

$\mathcal{N}$=2 $U(1)$, N_f=1 + mass, $(14;1,0;-5)$: $E_1=\mathbb{C}[\partial_{+\dot 1}\mathsf{c}\cdot x]$

$$x = \mathsf{b}^3\gamma_s - 3\beta^+\beta^-\mathsf{b}^2,\qquad d_1x\neq 0,\qquad \partial_{+\dot 1}\mathsf{c}\cdot Q_1x = Q_0w,\qquad d_2[\partial_{+\dot 1}\mathsf{c}\,x] = -\big[\partial_{+\dot 1}\mathsf{c}\,Q_2x - Q_1w\big]\neq 0$$

$$\text{quartic SQED: } x=\mathsf{b}^3\gamma_+\gamma_- - \tfrac{3}{2}\beta^+\beta^-\mathsf{b}^2;\qquad U(1)^2 \text{ triangle + mass}:\ a=\gamma_s,\ (12;0,0;-4)$$

8.6 Quantum lift

theory	$(\mathrm{wt};n_+,n_-;\mathrm{gh})$	flavour	$R;\ J_L,\ J_R$	d_r	rk	d_2 carried by
$\mathcal{N}$=2 $U(1)$, N_f=1 + mass	$(12;1,1;-4)$		2; 3, 0	d_2	1	$\langle a,Z,Y\rangle$
$\mathcal{N}$=2 $U(1)$, N_f=1 + mass	$(14;1,0;-5)$		2; 3, 1/2	d_2	1	$\mathcal{A}\cdot x$
$\mathcal{N}$=2 $U(1)$, N_f=1 + mass	$(16;1,1;-6)$		2; 4, 0	d_2	2	$\langle a,Z,Y\rangle + \mathsf{b}\,Y$
$\mathcal{N}$=2 $U(1)$, N_f=1 + mass	$(18;1,0;-7)$		2; 4, 1/2	d_2	1	$\mathcal{A}\cdot x$
$\mathcal{N}$=2 $U(1)$, N_f=1 + mass	$(20;1,1;-8)$		2; 5, 0	d_2	2	1 of 2
$\mathcal{N}$=2 $U(1)$, N_f=1 + mass	$(22;1,0;-9)$		2; 5, 1/2	d_2	1	$\mathcal{A}\cdot x$
$\mathcal{N}$=2 $U(1)$, N_f=1 + mass	$(24;1,1;-10)$		2; 6, 0	d_2	2	open
$U(1)$, quartic SQED	$(12;1,1;-4)$		2; 3, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)$, quartic SQED	$(14;1,0;-5)$		2; 3, 1/2	d_2	1	$\mathcal{A}\cdot x$
$U(1)$, quartic SQED	$(16;1,1;-6)$		2; 4, 0	d_2	2	$\langle a,Z,Y\rangle + \mathsf{b}\,Y$
$U(1)$, quartic SQED	$(18;1,0;-7)$		2; 4, 1/2	d_2	1	$\mathcal{A}\cdot x$
$U(1)$, cubic singlet	$(9;1,1;-4)$		2; 3, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)$, cubic singlet	$(12;1,1;-6)$		2; 4, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)$, cubic singlet	$(12;2,1;-5)$		3; 4, 1/2	d_2	1	$\mathcal{A}\cdot x$
$U(1)$, cubic singlet	$(15;1,1;-8)$		2; 5, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)$, cubic singlet	$(18;1,1;-10)$		2; 6, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)$, $\gamma_s^2\gamma_+\gamma_-$	$(11;1,1;-3)$	$(0,-1)$	5/2; 5/2, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)$, $\gamma_s^2\gamma_+\gamma_-$	$(11;2,1;-3)$	$(0,-1)$	5/2; 3, 1/2	d_2	1	1 of 2
$U(1)$, $\gamma_s^2\gamma_+\gamma_-$	$(12;1,1;-4)$		2; 3, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)$, $\gamma_s^2\gamma_+\gamma_-$	$(13;1,1;-4)$	$(0,1)$	5/2; 3, 0	d_2	1	$\mathcal{A}\cdot x$
$U(1)$, $\gamma_s^2\gamma_+\gamma_-$	$(13;2,1;-4)$	$(0,1)$	5/2; 7/2, 1/2	d_2	1	$\mathcal{A}\cdot x$
$U(1)$, $\gamma_s^2\gamma_+\gamma_-$	$(14;2,1;-4)$	$(0,-2)$	3; 7/2, 1/2	d_2	1	$\langle a,Z,Y\rangle$
$U(1)^2$ triangle	$(10;1,1;-4)$	$(1,1,1,1)$	8/3; 3, 0	d_2	1	open
$U(1)^2$ triangle	$(10;1,1;-4)$	$(-1,0,-1,0)$	8/3; 3, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)^2$ triangle	$(10;1,1;-4)$	$(0,-1,0,-1)$	8/3; 3, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)^2$ triangle + mass	$(12;0,0;-4)$		2; 2, 0	d_2	1	$\mathcal{A}\cdot x$
$U(1)^2$ triangle + mass	$(12;1,1;-3)$		3; 5/2, 0	d_2	1	$\mathcal{A}\cdot x$
$U(1)^2$ triangle + mass	$(12;1,1;-4)$	$(-1,0,-1)$	2; 3, 0	d_2	1	$\langle a,Z,Y\rangle$
$U(1)^2$ triangle + mass	$(14;1,1;-4)$	$(1,0,1)$	3; 3, 0	d_2	1	$\langle a,Z,Y\rangle$

theory	(wt; n_+, n_-; gh)	flavour	$R;\ J_L,\ J_R$	d_r	rk	d_2 carried by
$\mathcal{N}$=2 $U(1)$, N_f=1	$(11;1,1;-5)$	$(0,-1)$	$7/3;\ 7/2,\ 0$	d_3 [$\mathbb{Q}$]	1	–
$\mathcal{N}$=2 $U(1)$, N_f=1	$(12;1,1;-6)$		$2;\ 4,\ 0$	d_3	1	–
$\mathcal{N}$=2 $U(1)$, N_f=1	$(13;2,1;-6)$	$(0,-2)$	$8/3;\ 9/2,\ 1/2$	d_3	1	–
$\mathcal{N}$=2 $U(1)$, N_f=1	$(14;1,1;-7)$	$(0,-1)$	$7/3;\ 9/2,\ 0$	d_3	1	–
$\mathcal{N}$=2 $U(1)$, N_f=1	$(15;2,2;-7)$	$(0,-3)$	$3;\ 11/2,\ 0$	d_3	1	–
$\mathcal{N}$=2 $U(1)$, N_f=1	$(16;2,1;-8)$	$(0,-2)$	$8/3;\ 11/2,\ 1/2$	d_3	1	–
$\mathcal{N}$=2 $U(1)$, N_f=1	$(17;1,1;-9)$	$(0,-1)$	$7/3;\ 11/2,\ 0$	d_3	1	–
$U(1)$, cubic singlet	$(11;1,0;-5)$		$7/3;\ 3,\ 1/2$	d_3 [$\mathbb{Q}$]	1	–
$U(1)$, cubic singlet	$(13;1,1;-6)$		$8/3;\ 4,\ 0$	d_3	1	–
$U(1)$, cubic singlet	$(14;1,0;-7)$		$7/3;\ 4,\ 1/2$	d_3	1	–
$U(1)$, cubic singlet	$(17;1,0;-9)$		$7/3;\ 5,\ 1/2$	d_3	1	–
$U(1)$, quartic SQED	$(18;2,1;-6)$		$3;\ 9/2,\ 1/2$	d_3	2	–
$\mathcal{N}$=2 $U(1)^2$ quiver	$(11;1,1;-4)$	$(-1,1,1,1)$	$10/3;\ 3,\ 0$	d_2	1	open
$\mathcal{N}$=2 $U(1)^2$ quiver	$(11;1,1;-4)$	$(1,-1,-1,0)$	$10/3;\ 3,\ 0$	d_2	1	open

towers : $(\mathrm{wt};n_\pm;\mathrm{gh}) \to (\mathrm{wt}+kD;n_\pm;\mathrm{gh}-2k)$, same r $\quad$ (step $=(\mathrm{wt},\mathrm{gh})(\mathsf{b})$)

$\mathcal{N}$=2 $U(1)$, N_f=1, 2 : $\mathrm{rk}\, d_2 = 0$ on 566 blocks with $\ker d_1 \neq 0$ (311 with $Q_2\mathcal{O}_0 \neq 0$); $\quad \mathrm{rk}\, d_3 \neq 0$ (N_f=1)

$\mathcal{N}$=2 $U(1)^2$ quiver : $\mathrm{rk}\, d_2 = 1$ at $(11;1,1;-4)_{(-1,1,1,1)}$, $(11;1,1;-4)_{(1,-1,-1,0)}$

$\mathrm{rk}\, d_4 = 0$ on 1072 tested blocks (67 with nonzero target); $\quad Q_4 \to -Q_4,\ 2Q_4$ at $(12;1,1;-6) \ \Rightarrow\ \mathrm{rk}\, d_4 = 1$

Q_4 (fans, $\deg \mathcal{W} \le 3$) : $\displaystyle\sum_{j=0}^{4} Q_j Q_{4-j} = 0$ on all tested words; violated at $-Q_4,\ 2Q_4$

Physical identification

$\mathcal{N}$=2 $SU(2)$, monopole point : $\sqrt{2}A_D M\tilde{M} \ \leftrightarrow\ \gamma_s\gamma_+\gamma_-$

$\Longrightarrow$ the d_3 lifts of $\mathcal{N}$=2 $U(1)$, N_f=1 are lifts of this low-energy theory

8.7 Data

Files `data/abelian_census.csv`, `data/abelian_primary_lifts.csv`

9. Non-abelian components: $G = SU(2),\ SU(3)$

9.1 Action and components

$SU(N)$ with an adjoint chiral Φ and a singlet S $\quad A = 1,\dots,N^2-1$; t_A Chevalley basis of $\mathfrak{sl}_N$ in the fundamental

$$S_{\mathrm{tw},E} = \int_{\mathbb{C}^2} \Omega \wedge \Big[\mathsf{b}_A\big(\bar\partial \mathsf{c}^A + \tfrac{1}{2} f^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big) + \big(\bar\partial(\gamma_\Phi)^A + f^A{}_{BC}\mathsf{c}^B(\gamma_\Phi)^C\big)(\beta_\Phi)_A + \bar\partial\gamma_S\,\beta_S + \mathcal{W}(\gamma) - \tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}\Big]_{(0,2)}$$

$$[t_B, t_C] = f^A{}_{BC} t_A, \qquad \mathsf{w}^A_{\dot a} \mapsto \tfrac{1}{2}\partial_{+\dot a}\mathsf{c}^A$$

$$\mathcal{I} = \mathcal{W}(\gamma) + f^A{}_{BC}(\beta_\Phi)_A\mathsf{c}^B(\gamma_\Phi)^C + \tfrac{1}{2} f^C{}_{AB}\,\mathsf{b}_C\mathsf{c}^A\mathsf{c}^B - \tfrac{1}{4}\mathsf{w}^A_{\dot a}\mathsf{w}^{A\dot a}$$

vertices = monomials of $\mathcal{I}$, $\qquad Q_0 X = \dfrac{\partial_L \mathcal{I}}{\partial(\text{partner of } X)}$, $\qquad$ partners: $(\gamma,\beta),\ (\mathsf{c}^A, \mathsf{b}_A)$

theory	matter	$\mathcal{W}$	weights, D
$\mathcal{N}$=2 $SU(N)$	Φ adjoint	0	$w_\Phi = 1,\ D = 3$
$SU(2)$ + flip singlet	Φ adjoint, S singlet	$\frac{1}{2}S\,\mathrm{tr}\,\Phi^2 = S\big((\gamma^H_\Phi)^2 + \gamma^E_\Phi\gamma^F_\Phi\big)$	$w_\Phi = w_S = 1,\ D = 3$

$\mathcal{I} = \mathcal{W} + \sum q\beta\mathsf{c}\gamma$ reproduces the $U(1)^k$ rules of §8 ($Q_0 \dots Q_3$)

9.2 Lie algebra realization

$SU(2)$: (E,H,F), $\quad [H,E] = 2E,\ [H,F] = -2F,\ [E,F] = H$; $\qquad SU(3)$: $\{E_{ij}\}_{i\neq j} \cup \{E_{11}-E_{22}, E_{22}-E_{33}\}$

$\rho_{\mathrm{adj}}(t_A)^C{}_B = f^C{}_{AB}$, $\qquad$ gauge action on letters: $t_A\gamma_i = \rho(t_A)_{ij}\gamma_j$, $\quad t_A\beta^j = -\rho(t_A)_{ij}\beta^i$

9.3 Invariant blocks

$$\text{block } (\mathrm{wt}; n_+, n_-; \mathrm{gh})_f, \quad f = \#\gamma_\Phi - \#\beta_\Phi - 2(\#\gamma_S - \#\beta_S); \qquad C_{\mathrm{inv}} = \{\text{Cartan weight } 0\} \cap \bigcap_k \ker E_{k,k+1}$$

$$[E_{k,k+1}, Q_n] = 0 \ (n \le 3), \qquad Q_0^2 = 0, \qquad \sum_{a=0}^{n} Q_a Q_{n-a} = 0 \ (n = 2, 3) \qquad (\text{checked on random words, } SU(2), SU(3))$$

$$SU(2) \text{ flip}: \ R = \tfrac{2}{3}\mathrm{wt} + \mathrm{gh}, \qquad J_L = \tfrac{1}{2}(n_+ + n_- - \mathrm{gh}), \qquad J_R = \tfrac{1}{2}(n_+ - n_-)$$

9.4 Frontier

(d_2, d_3 on every block with $\ker d_1 \neq 0$; d_4 at wt $= 13, 14$)

theory	wt	$J = n_+ + n_-$	blocks with $E_1 \neq 0$	$\sum E_1$	$\sum \mathrm{rk}\, d_1$	$\sum \mathrm{rk}\, d_2$	$\sum \mathrm{rk}\, d_3$	$\sum \mathrm{rk}\, d_4$
$\mathcal{N}$=2 $SU(2)$	≤ 12	≤ 2	152	432	263	7	0	–
	13, 14 (partial)	≤ 2	31	68	41	10	0	0
$SU(2)$ + flip singlet	≤ 12	≤ 2	176	503	334	7	0	–
	13, 14 (partial)	≤ 2	31	55	48	0	1	0
$\mathcal{N}$=2 $SU(3)$	8–12 (partial)	≤ 1	31	51	34	2	0	–

9.5 Classes

$$L = \mathsf{b}_A(\gamma_\Phi)^A \quad (4; 0, 0; -2)_1, \qquad d_1 L \neq 0$$

$$M_1 = (\beta_\Phi)_E(\beta_\Phi)_F(\beta_\Phi)_H \quad (6; 0, 0; -3)_{-3}, \qquad J_f = \beta_S\gamma_S - \tfrac{1}{2}(\beta_\Phi)_A(\gamma_\Phi)^A \quad (3; 0, 0; -1)_0$$

$$Q_1 M_1 = 0, \qquad M_1\, Q_1 L \in \mathrm{im}\, Q_0, \qquad Q_1(LM_1) - (Q_1 L)M_1 \in \mathrm{im}\, Q_0 \implies LM_1 \in E_2$$

9.6 Quantum lift

theory	$(\mathrm{wt}; n_+, n_-; \mathrm{gh})_f$	$R;\ J_L,\ J_R$	E_1	$\mathrm{rk}\, d_1$	d_r	rk	class	
$\mathcal{N}$=2 $SU(2)$	$(10; 0, 0; -5)_{-2}$	5/2, 0	1	0	d_2	1	$L\,M_1$	[$\mathbb{Q}$]
	$(10; 1, 0; -5)_{-2}$	3, 1/2	3	1	d_2	1		[p]
	$(10; 1, 1; -5)_{-2}$	7/2, 0	9	4	d_2	1		[p]
	$(11; 0, 0; -6)_{-1}$	3, 0	1	0	d_2	1	$L \cdot E_1(7; 0, 0; -4)_{-2}$	[p]
	$(11; 1, 0; -6)_{-1}$	7/2, 1/2	2	1	d_2	1		[p]
	$(11; 1, 1; -6)_{-1}$	4, 0	4	3	d_2	1		[p]
	$(12; 1, 1; -5)_0$	7/2, 0	1	0	d_2	1		[p]
	$(13; 1, 1; -6)_{-2}$	4, 0	8	1	d_2	3		[p]
	$(13; 1, 1; -6)_1$	4, 0	1	0	d_2	1		[p]
	$(14; 0, 0; -7)_{-1}$	7/2, 0	1	0	d_2	1		[p]
	$(14; 1, 0; -7)_{-1}$	4, 1/2	3	1	d_2	1		[p]
	$(14; 1, 1; -7)_{-1}$	9/2, 0	13	6	d_2	4		[p]
$SU(2)$ + flip singlet	$(9; 1, 1; -3)_0$	3; 5/2, 0	5	4	d_2	1	$J_f \cdot E_1(6; 1, 1; -2)_0$	[$\mathbb{Q}$]
	$(11; 0, 0; -5)_2$	7/3; 5/2, 0	1	0	d_2	1	$L \cdot E_1(7; 0, 0; -3)_1$	[p]
	$(11; 1, 0; -5)_2$	7/3; 3, 1/2	2	0	d_2	1		[p]
	$(11; 1, 1; -5)_2$	7/3; 7/2, 0	6	2	d_2	1		[p]
	$(12; 0, 0; -6)_3$	2; 3, 0	1	0	d_2	1	$L \cdot E_1(8; 0, 0; -4)_2$	[p]
	$(12; 1, 0; -6)_3$	2; 7/2, 1/2	1	0	d_2	1		[p]
	$(12; 1, 1; -6)_3$	2; 4, 0	2	1	d_2	1		[p]
	$(14; 1, 0; -6)_2$	10/3; 7/2, 1/2	1	0	d_3	1	$X\,Y$	[$\mathbb{Q}$, cert]
$\mathcal{N}$=2 $SU(3)$	$(11; 0, 0; -5)_{-1}$	5/2, 0	3	2	d_2	1		[p]
	$(12; 0, 0; -6)_0$	3, 0	2	1	d_2	1		[p]

$$X \in E_1(11; 0, 0; -5)_2, \qquad Y \in E_1(3; 1, 0; -1)_0; \qquad \mathcal{N}\text{=2 } SU(N): \ \mathcal{W} = 0, \ R \text{ omitted}$$

$$\mathcal{N}\text{=2 } SU(3): \ (10; 0, 0; -5)_{-2} \text{ has } E_1 = 2, \ \mathrm{rk}\, d_1 = 2$$

$$[\mathbb{Q}, \text{cert}]: \ \text{chain } (x_0, x_1, x_2), \ \phi = (\phi_1, \phi_2, \phi_3) \text{ over } \mathbb{Q};$$

$$Q_0 x_0 = 0, \quad Q_1 x_0 + Q_0 x_1 = 0, \quad Q_2 x_0 + Q_1 x_1 + Q_0 x_2 = 0, \quad E\, x_k = 0$$

$$\sum_{k=m}^{3} \phi_k\big(Q_{k-m} y\big) = 0 \ \ \forall \text{ monomials } y \text{ of stage } m = 1, 2, 3 \ (H = 0;\ 16400,\ 64038,\ 54429)$$

$$\phi_3\big(Q_3 x_0 + Q_2 x_1 + Q_1 x_2\big) = 1$$

$$\#\text{terms}: \ x_{0,1,2} = 208,\ 9535,\ 43941; \quad \phi_{1,2,3} = 11786,\ 5350,\ 501$$

9.7 Data

Files `data/su2_lifts.csv`

10. Running

item	computation
$SO(7)$ vs $Sp(3)$	E_2, d_2, d_3, all sectors, $E + J_L = \frac{23}{2}, 12$; E_2 per (E, R_I, J_L, J_R)
$SO(9)$ vs $Sp(4)$	same, $E + J_L \leq 12$ (all sectors)
G_2	h, ρ, d_1, d_2, d_3, all sectors, $E + J_L \leq 15$; trace words vs exact singlet counts
F_4, E_6, E_7	trace-word span vs $\dim \mathrm{Hom}_G(\mathrm{adj}^{\otimes k}, \mathbb{C})$
$SO(7)$ β	d_1 from $(7, 2, 2; 1, 1)Y10$ (recheck), chains
SO/Sp scans	$Sp(3..5)$, $SO(11)$: $E + J_L \in [\frac{25}{2}, 14]$
$SU(4)$, $SU(5)$	last shapes at $E + J_L = 14$, $\frac{29}{2}$
quiver, SCQCD	$SU(2)^2$ to $E + J_L = 14$, $SU(3)^2$ to 10, SCQCD $SU(2)$ 12; d_3 scans
massive flows	w scans (all five theories); $SU(3)$ one mass $(7, 7; 1, 0; -4)$ d_2
$SU(3)$ β, LS	$E + J_L \in [\frac{19}{2}, 11]$, d_2, d_3; operator form of the lifted pairs
$SU(3)$ LS	$E + J_L = 12$, $\frac{25}{2}$: d_1, d_2, d_3 census
abelian	cubic singlet wt = 19–24, $U(1)^2$ triangle wt = 13, 14: d_2, d_3, d_4; d_5 targets $\mathcal{N}$=2 $U(1)$, N_f=3; $\mathcal{N}$=2 $U(1)^2$ quiver: d_2, d_3
$SU(2)$, $SU(3)$ components	$\mathcal{N}$=2 $SU(2)$, flip: wt = 13, 14, $J \leq 2$, d_2, d_3, d_4; $\mathcal{N}$=2 $SU(3)$: wt ≤ 12, $J \leq 1$